

CNC DRILLING MACHINE

SKH-690 / SKH-612H / SKH-612S

MANUAL

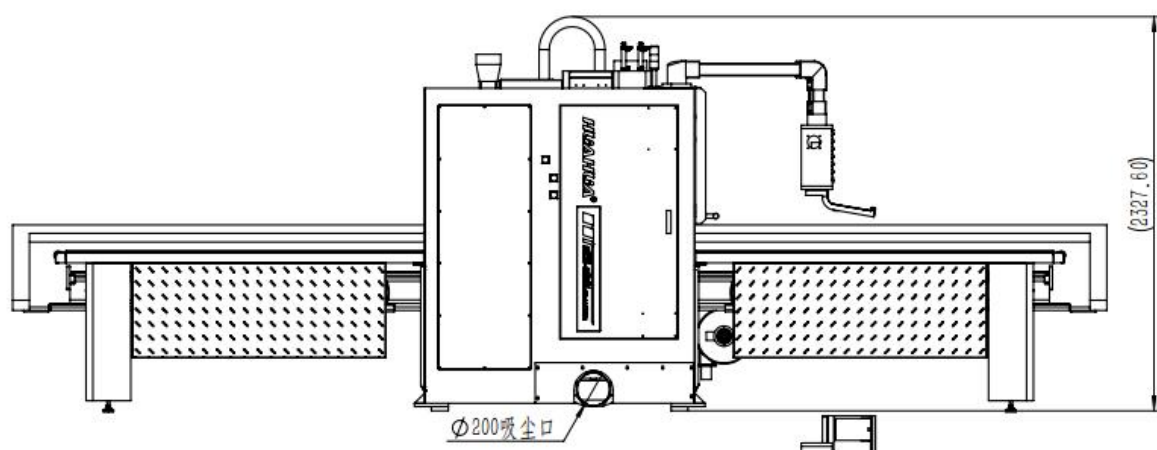


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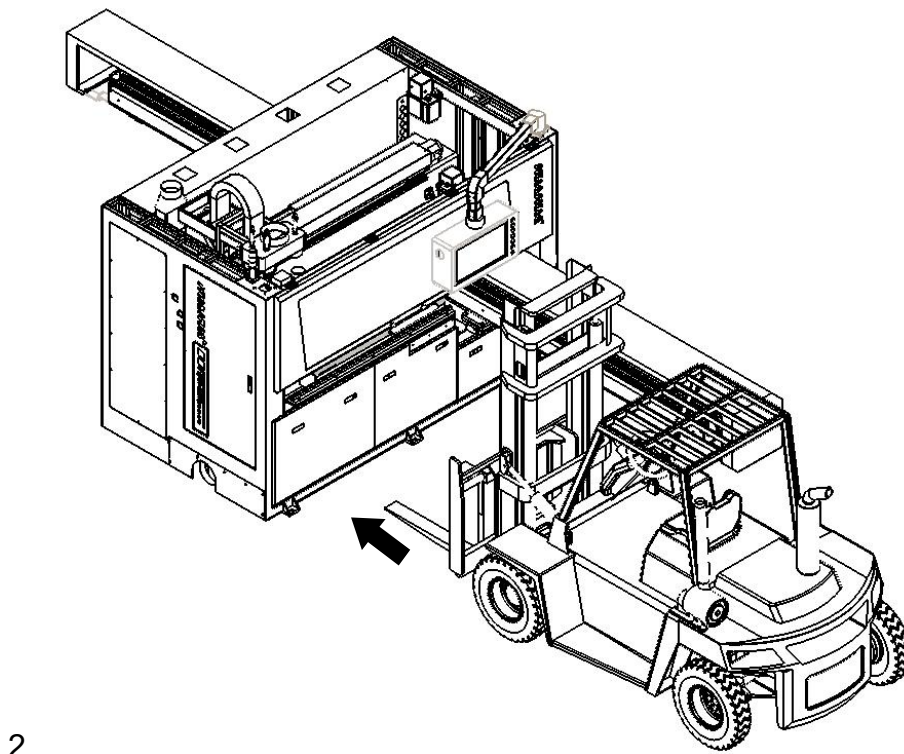
1. Introduction to the machine

This equipment is mainly used for drilling, grooving, cutting, milling and other functions of solid wood particle board, fiberboard, sandwich board and plywood. It is a machine equipped with a program control system. Please do not use it for other purposes than the original setting.

This equipment adopts Taiwan's new generation digital control system, and the system integrates CAM software, which can scan the bar code or edit the graphics and generate the processing program. The working procedure is highly concentrated. One clamping can complete the drilling of six sides of the workpiece, as well as the grooving and milling of the upper and lower parts of the workpiece.

According to the user's production needs, you can choose different discharging methods (forward forward, forward backward), and also choose automatic conveyor roller or side discharge, which is convenient for connection or multi-machine side-by-side production.

2. Installation precautions



2.

Picture 2

2.1 Handling precautions

- (1) When carrying, keep the center of gravity balanced to prevent the machine from tilting.
- (2) When using a motorized forklift, make the center of gravity of the machine face backward.
- (3) Only qualified personnel can set up a motorized forklift.
- (4) It should be marked with words such as "Fragile, do not press and do not touch", and you should be especially careful to avoid damage to its parts and components, and handle with care to avoid violent vibration or collision.

2.2 Packing

Transport and hoisting size reference diagram

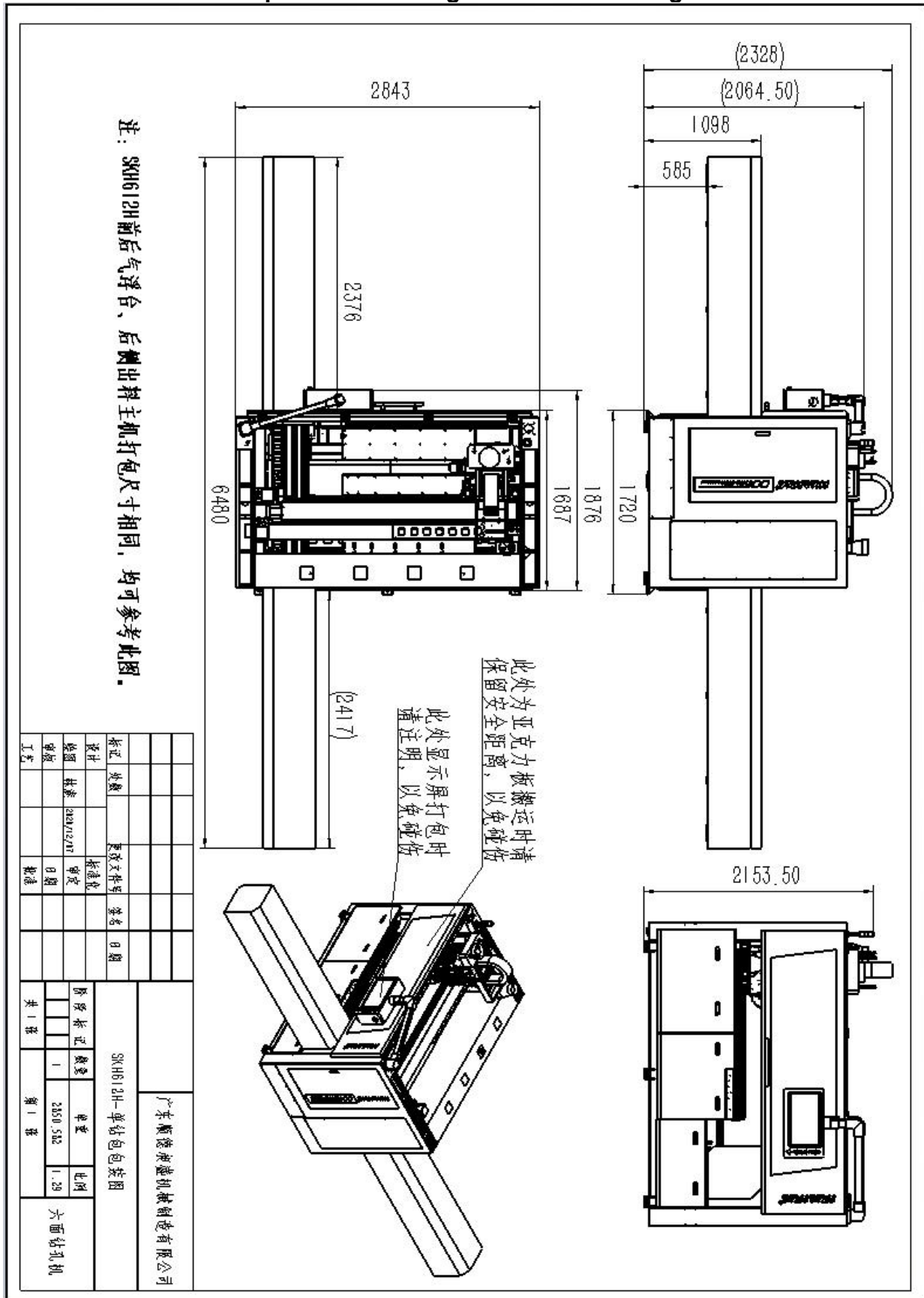


图1

2.3 Set up and debug the machine

2.3.1 Precautions

- (1) The machine installation must be installed by professional and technical personnel.
- (2) Before installing the machine, check whether the parts of the machine are in good condition.
- (3) In order to allow users to clean, maintain, and open the electrical box conveniently and easily, when the machine and the machine, or with the wall, and other objects are placed together, each side of the machine must be kept at least 1000mm Above distance
- (4) The machine cannot be directly placed on the floor, and the horizontal height must be adjusted with adjusting screws and spacers. The floor of the use site must be a flat concrete floor, and do not place it on an unstable foundation, otherwise it will cause mechanical imbalance.
- (5) It is recommended that a certain safety distance be reserved around the working area of the equipment to avoid injuries such as collisions caused by equipment operation.

2.3.2 Installation accuracy adjustment

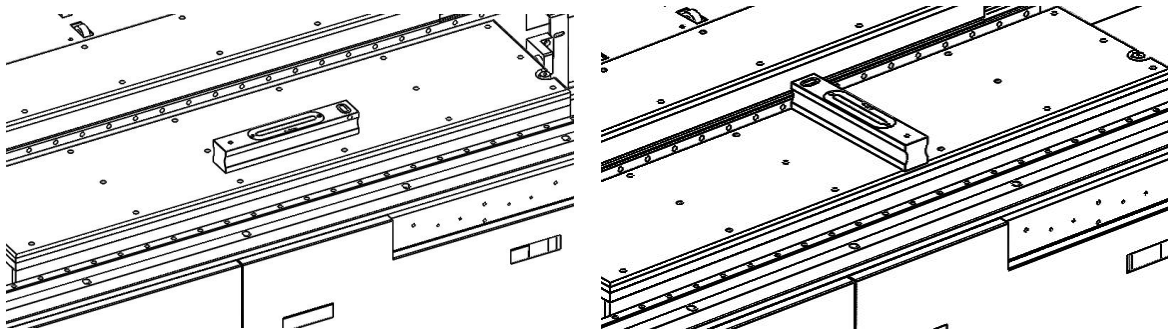
(1) Adjust the machine level:

The following tools are needed:

- 4 hexagon head screws M16*60L;
- Mat;
- 2 pieces of level gauges, please use the level gauge with an accuracy of 0.02mm/M or more;
- Toolbox wrench;

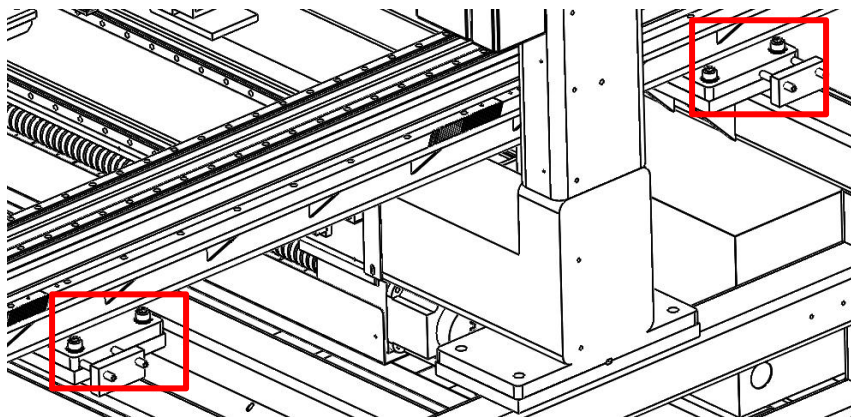
Steps:

- Install the 4 screws on the screw holes of the base of the machine foot, and put the cushion between the ground and the bottom of the screws;
- Clean the table first, and place the spirit level in both the horizontal and vertical directions, as shown in the figure below:

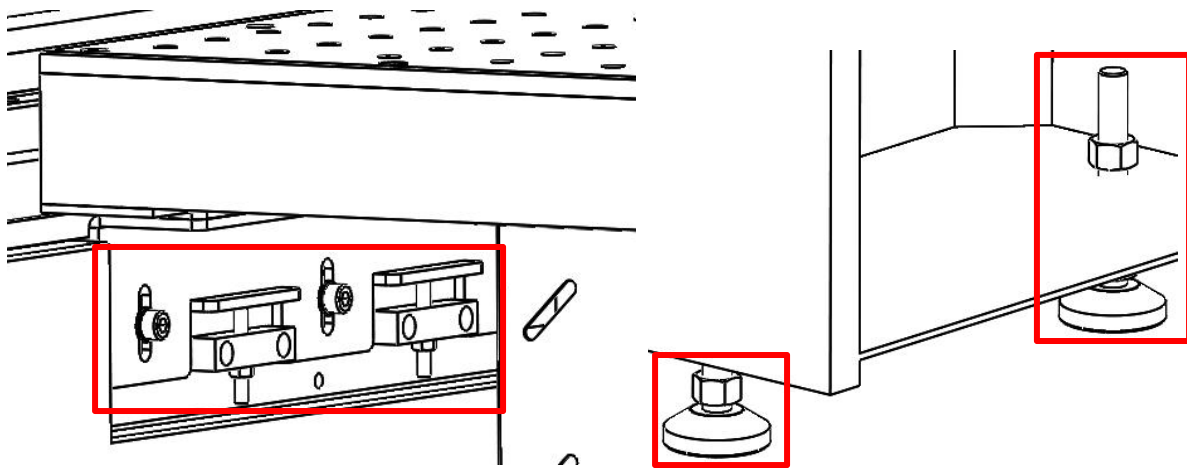


- Turn the hexagon head screw to adjust to the level indicated by the spirit level, and the adjustment accuracy should be kept within $\pm 0.04\text{mm}$.

Feeding beam adjustment: first clean the feed beam base surface and the rack mounting surface, place the feed beam assembly on the rack mounting surface, and push the feed beam assembly to the right so that the two bases of the feeding beam are close to the positioning screw on the right side, and lock the front and back Screw the base, and install the drag chain groove and drag chain components of the manipulator, as shown in the figure. (This section only exports equipment, you must operate according to this item)

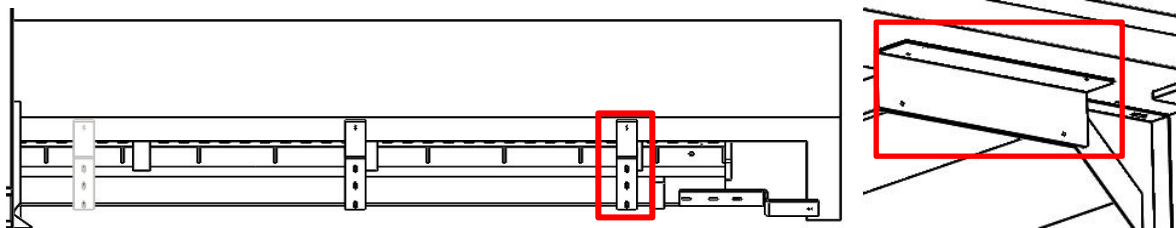


(1) Air-floating table adjustment: install the air-floating table to the base frame, place the level on the air-floating table, turn the adjusting studs and anchor bolts so that the air-floating table is parallel to the rack table and 0.1 higher than the rack table mm.



(1) Installation of front and rear guards of the feeding beam: Install the guard connecting plate to the feeding beam, lift up the front and rear guards and place them on the connecting plate, and tighten the screws.

Note: The installation method of front and rear shields is the same, hereafter the shields are taken as an example



(1) Before adjusting the origin of the machine, remove the packing fixing screws of the upper and lower drill packs. After adjusting the accuracy of the machine, it is necessary to set the origin again.

2.3.3 Power connection

- (1) All wiring connections must be performed by qualified personnel;
- (2) Make sure that the power supply voltage matches the machine voltage;
- (3) The main power line should be connected to the terminal of R.S.T in the electrical box;
- (4) Use 1 wire with a single wire diameter of at least 4mm²;
- (5) It is forbidden to perform electrical operations blindly, and not to change the operating position at will;
- (6) The equipment must be grounded, and must not be connected to the ground with other machinery, because if other machinery is exposed to electricity, it will backwash to the computer host and cause irreparable damage.

Note: The grounding wire must comply with the regulations! (The depth is greater than 1.5 meters)

2.3.4 Install the equipment for the first time and check before use

- (1) Fast interface of all circuits of the equipment, aviation connector, VGA, data line and other interfaces;
- (2) Air pressure and cylinder feed speed check (press plate, press wheel, manipulator, Z-axis auxiliary cylinder);
- (3) Check the pressure plate, pressure wheel, manipulator signal light;
- (4) X-axis, Y-axis, Z-axis, U-axis, V-axis, W-axis, A-axis, horizontal tool*4, positioning column, mechanical hand photoelectric proximity switch lock tightness inspection;
- (5) Please use a vacuum cleaner before using the equipment

3. Safety precautions

3.1. Operation qualification

Operators and maintenance personnel of the machine must receive professional training with the machine in advance, and fully grasp the operating procedures and safe operation procedures. For those who have not been trained in the system, it is strictly forbidden to operate on the machine alone. Be sure to implement this clause as a company regulation.

3.2. Management system

- (1) Set up a dedicated person to be responsible for safety, and regularly conduct safety education for operators to ensure the safety of the working environment, and strive to improve and raise the awareness of safe operation;
- (2) In the event of an accident, please turn off the machine immediately, find out and record the cause of the accident, and contact our company. The machine is equipped with a safety emergency stop switch, please pay attention to its location and keep it in mind.

3.3. Hazardous area

Please do not put your body into the movable part of the machine. When entering this part for maintenance of the machine, be sure to cut off the power in advance in accordance with the operating regulations and confirm that the part has stopped before proceeding. Otherwise, the operation is prohibited.

3.4. Precautions for operation

- (1) Please do not modify the shell (cover), control switch and control circuit;
- (2) The drilling size must be absolutely controlled within the machine's capacity;
- (3) Please do not open the cover (cover, shell, electric box) door, etc. to operate.

3.5. Precautions during maintenance

When performing maintenance, be sure to cut off the power supply in accordance with the operating regulations, and the workshop power supply should also be cut off. In order to prevent others from turning on the power by mistake, please show a warning sign with (non-professionals prohibited operation).

3.6 Warning signs

Depending on the structure of the machine, the following lists the danger signs that may appear on the machine

	Do not rely on
	DANGER
	Watch out for your hands
	Beware of electric shock

4.The main technical parameters of the machine

4.1. Technical parameters

Item	project name	parameter	Machine model	
			SKH690H	SKH612H
1	Each axis stroke/ Maximum positioning speed	X/U axis	5800mm&130m/min	5800mm&130m/min
		Y axis	1380mm&80m/min	1680mm&80m/min
		Z axis	90mm&30m/min	90mm&30m/min
		V axis	900mm&80m/min	1200mm&80m/min
		W axis	100mm&30m/min	100mm&30m/min
		A axis	960mm&30m/min	1260mm&30m/min
2	Drilling combination Machine	Vertical drill	Up drilling box: 12pcs Bottom drilling bo: 9pcs	Up drilling box: 12pcs Bottom drilling bo:9pcs
		Horizontal drill	8pcs	8pcs
		Motor speed of drill set	2800rpm(r/min)	2800rpm(r/min)
		Motor Power	2.2KW	2.2KW
3	Engraving and milling spindle*2	Spindle power	3.5KW*2	3.5KW*2
		Shank diameter	1/2	1/2
		Spindle form	Er25	Er25
		Spindle speed	18000rpm	18000rpm
4	Processing size range	Maximum width of panel	900mm	1200mm
		Minimum width of panel	50mm	50mm
		Maximum panel length	5000mm	5000mm
		Minimum length of panel	250mm	250mm
		Maximum thickness of panel	48mm	48mm
		Minimum thickness of panel	10mm	10mm
5	Other parameters	Height of worktable	960mm	960mm
		Total power	19.2KW	19.2KW
		Operating Voltage	380V/50Hz	380V/50Hz
		working pressure	0.7-0.8Mpa	0.7-0.8Mpa
		Machine net weight	3000kg	3500kg
		Machine dimensions	6500x2540x2200mm	6500x2840x2200mm
6	Six-sided drill line	Normal processing smallest board	120x300mm	120x300mm

4.2. Machine nameplate

HUAHUA — 桦桦数控 —	CNC DRILLING MACHINE 六面钻铣加工中心	
Model No.: SKH-612H	加工尺寸范围/Processing Size Range:	
钻孔组合装置/Drilling Unit: 垂直钻 15X方向+16Y方向 Vertical Spindle 15(X axis)+16(Y axis)	板材最小宽度 Min.width of work-piece 50mm 板材最大宽度 Max.width of work-piece 1220mm 板材最小长度 Min.length of work-piece 250mm 板材最大厚度 Max.thickness of work-piece 50mm	
水平钻 2左+2右及2前+2后 Horizontal Spindle 2(Left)+2(Right) and 2(Front)+2(Back) 钻组电机功率 Motor Power 2.2KW, 2800rpm	其他参数/Other Data: 工作台高度 Working table height 960mm 总功率 Total Power 19.2KW 工作电压 Working Voltage 380V 50Hz 工作气压 Air pressure 0.7-0.8Mpa 机器净重 Weight 3900KG 机器外形尺寸 Overall Dimensions 6480x3010x2200mm	
镗铣加工主轴/Router: 主轴形式 Router Type Er25 刀柄直径 Tool Shank Diameter $\varnothing 1\frac{1}{2}$ " 主轴电机功率 Router Power 3.5KW 主轴转速 Rotation Speed 18000rpm	出厂编号 Serial No.: 出厂日期 Out Factory Date	
广东顺德昶盛机械制造有限公司 Guangdong Shunde Changsheng Machinery Manufacturing Co., Ltd. 地址: 广东省佛山市顺德区伦教镇熹涌工业区 Longwei Road, Xichong Industrial Park, Lunjiao, Shunde District, Foshan, Guangdong, China		电话/Tel.: +86-0757-27336257 传真/Fax: +86-0757-28798598 Made in China

A: Machine name

B: Machine model

C: The relevant parameters of the upper and lower drilling package combination

Machine

D: Related parameters of router spindle

E: Equipment factory number

F: company name

G: company address

H: Equipment processing size range

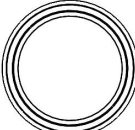
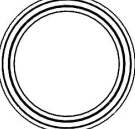
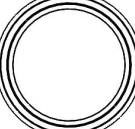
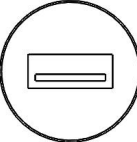
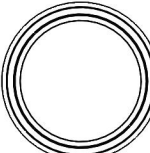
I: Other main parameters of the equipment

J: Date when the Machine was shipped

K: company contact number

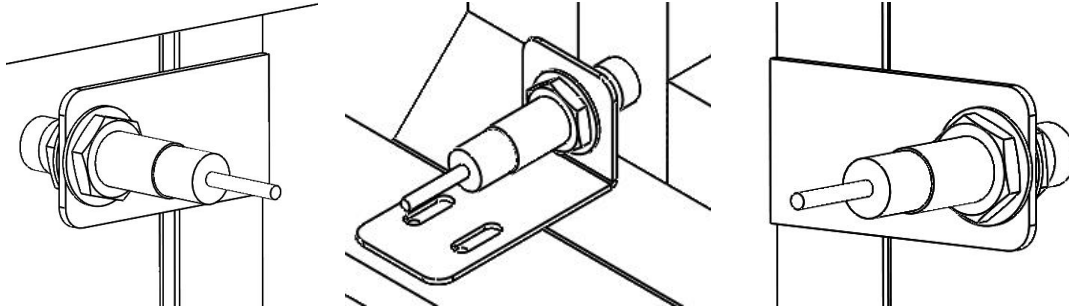
L: company fax

4.3. Operation Console button switch function

Picture	Name	Function Description
	急停 EMERGENCY	When it is found that the machine is running abnormally or an emergency situation occurs, you can press this button, all actions will stop, and an "emergency stop" fault will be reported. You must release this button when you restart the machine, that is, you only need to rotate it clockwise about 45° release
	程序启动 PROGRAM STARTUP	When the equipment is in the state of single block stop (M00), press this button to start the next processing execution action
	程序暂停 PROGRAM HOLD	When the equipment is in the "processing" state, press this button to pause to the current work action
	电源启动 POWER INDICATOR	Light On: Indicates that the system is powered on and ready to run. No light On: indicates that the system has not been powered on.
	电源停止 POWER OFF	Turn off the power supply of the machine
	USB	Users can transfer data through external machines (such as U disk, etc.)
	电脑启动 COMPUTER START	

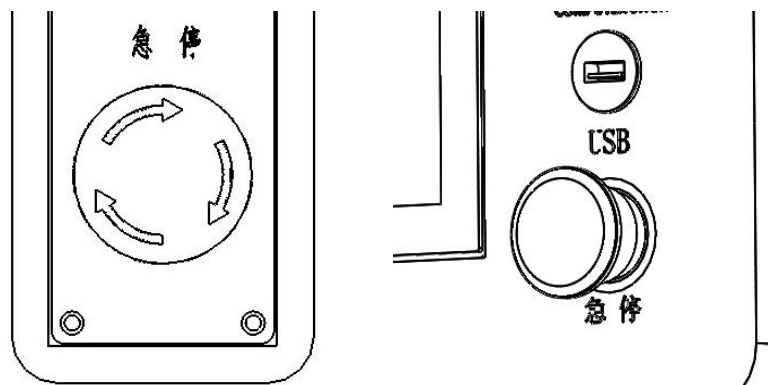
4.4 Safety machine

4.4.1. Proximity switch



The above figure shows the proximity switch. When the equipment is operating normally, open the left and right doors and the front guard door to stop the equipment.

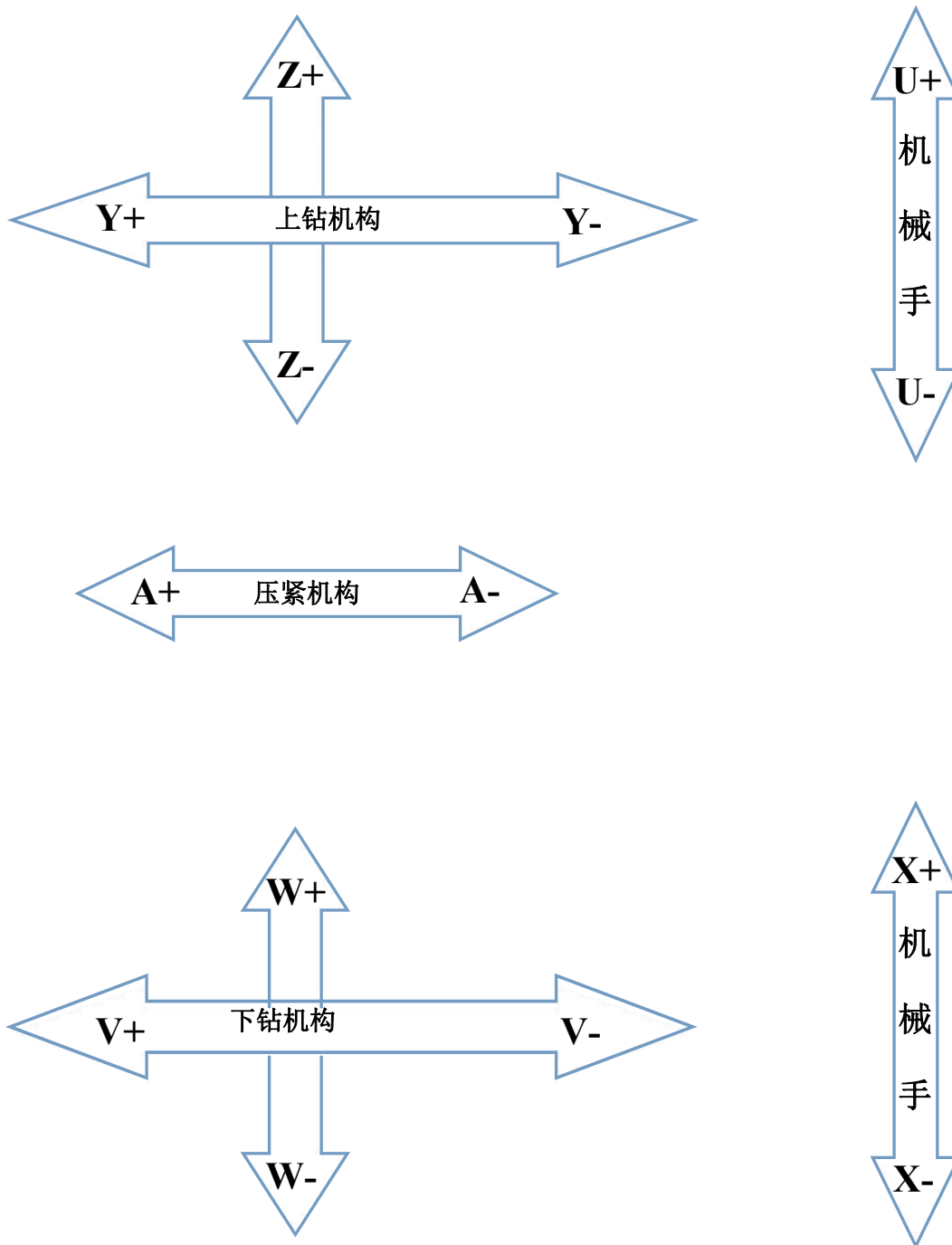
4.4.2. Emergency stop switch



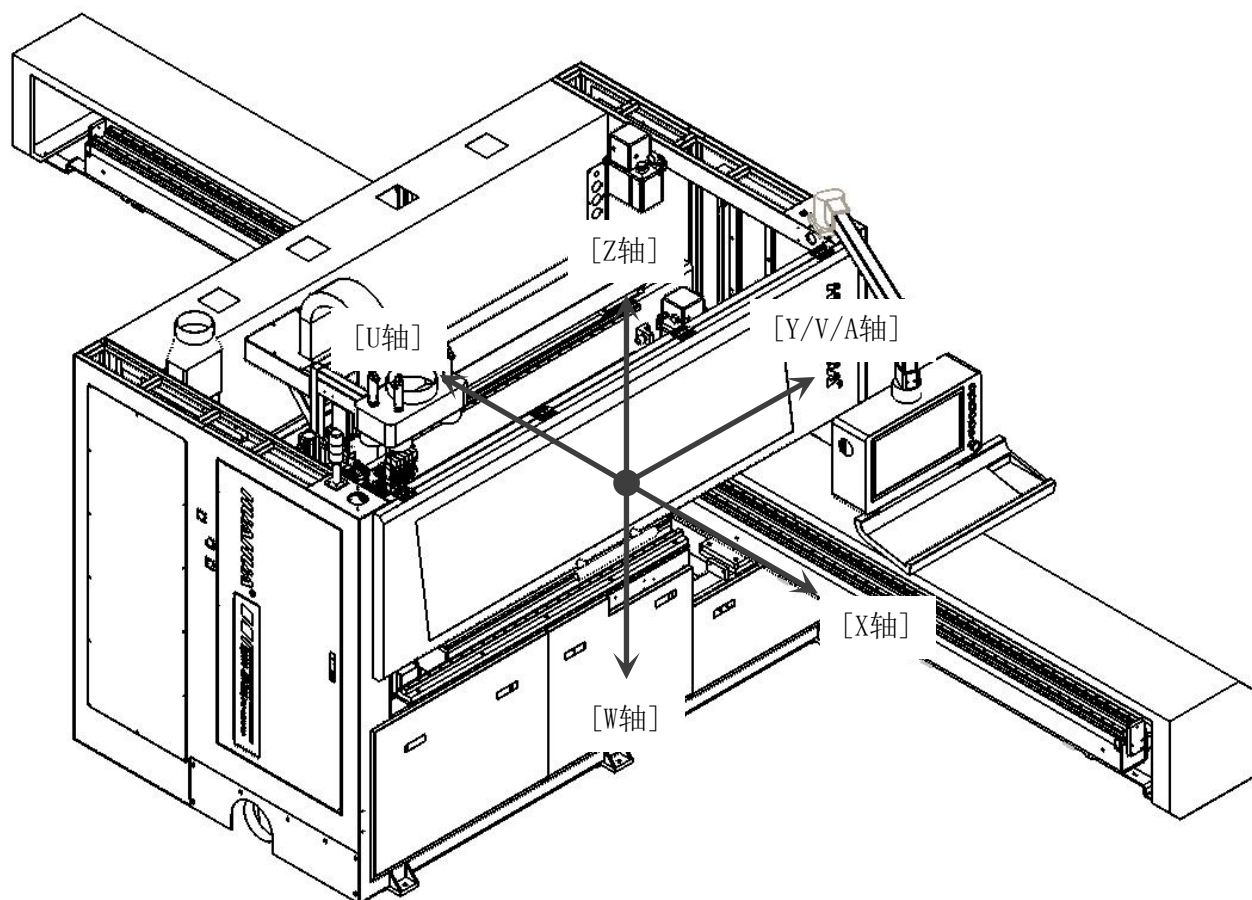
When it is found that the machine is running abnormally or an emergency occurs, you can press this button to stop all actions and report an "emergency stop" fault.

5. Technical Information

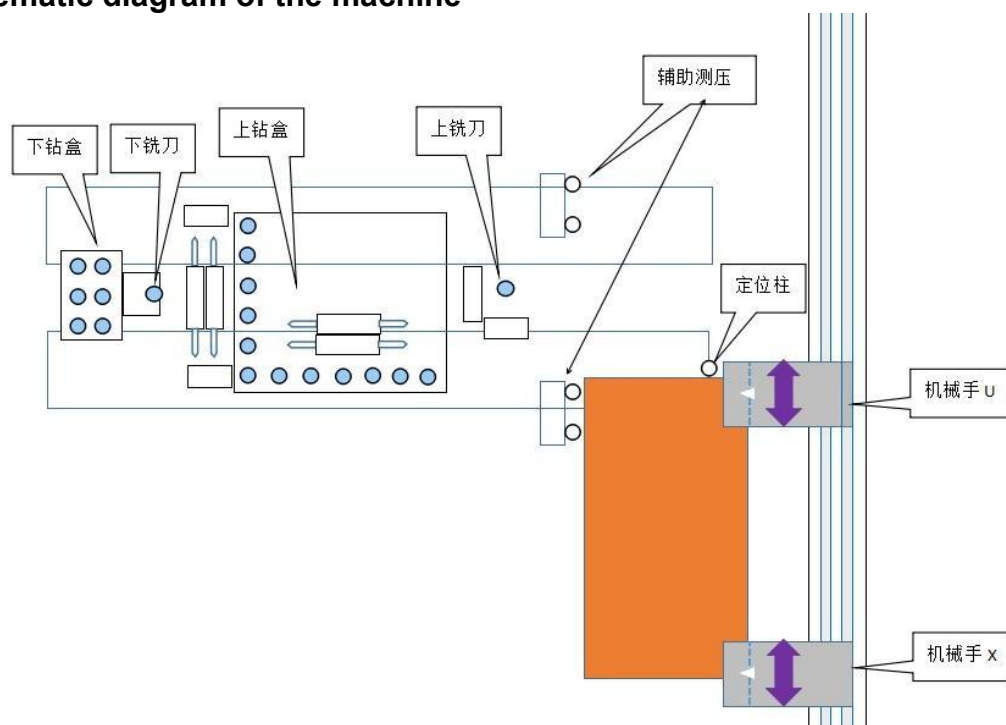
5.1. Machine coordinate system



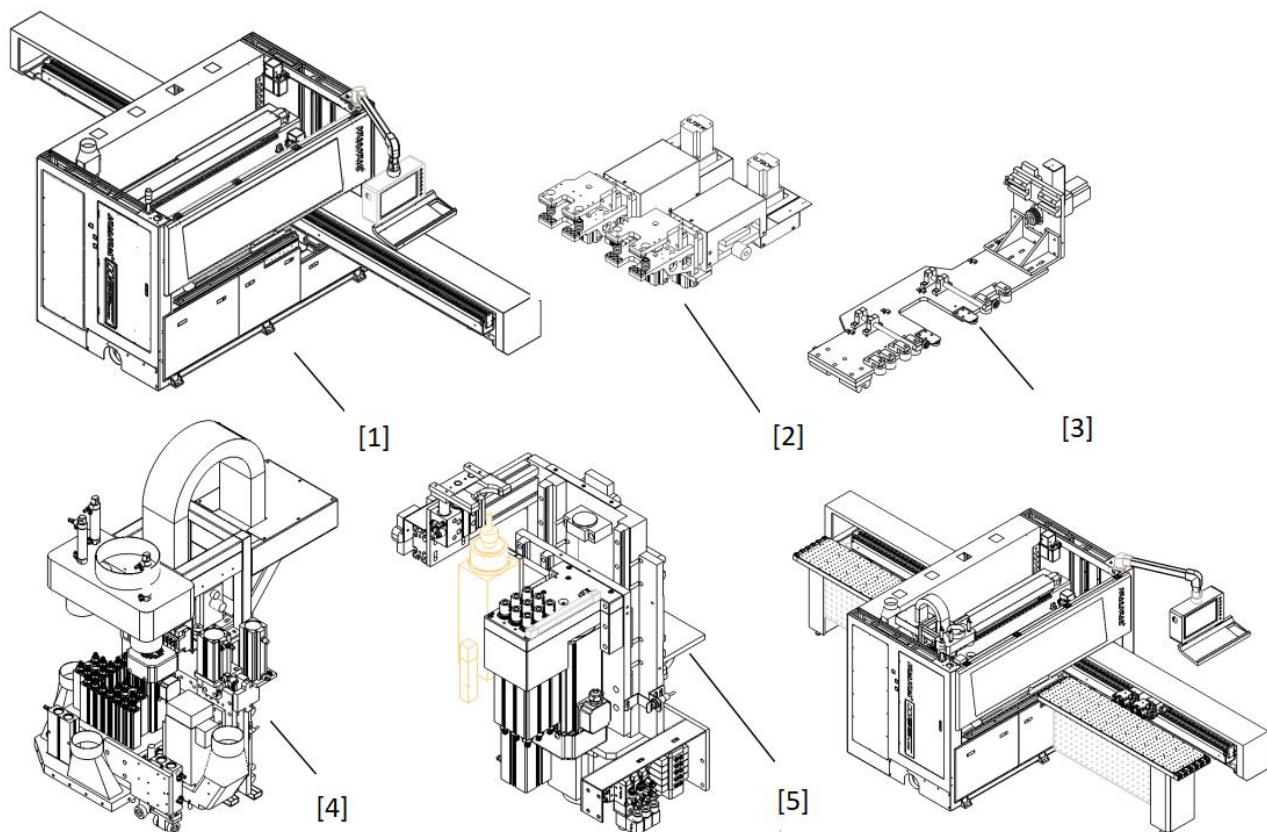
5.2. Schematic diagram of each axis coordinate



5.3. Schematic diagram of the machine

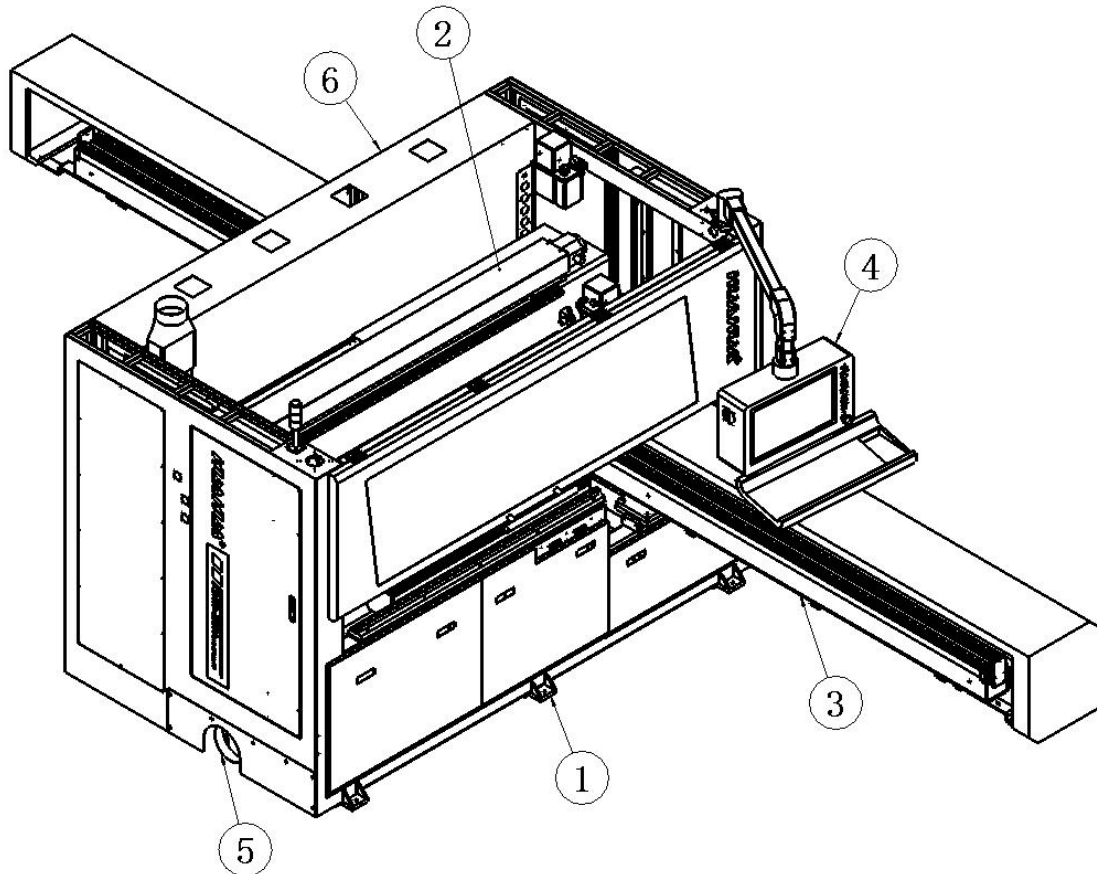


6. Introduction of working machine



Item #s	name	Qty
1	Rack components	1
2	double clamps assembly	1
3	Side aligning component	1
4	Up drilling box	1
5	Bottom drilling box	1

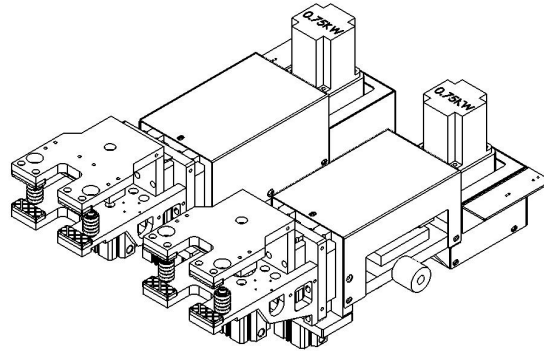
6.1. Rack components



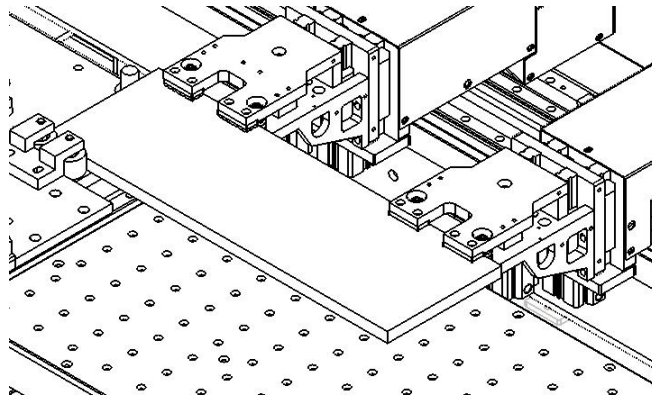
Item #s	name
1	Rack base assembly
2	Beam components
3	Feed beam assembly
4	Operation screen components
5	Dust removal components
6	Electric box components

Working principle: The rack components are composed of rack base components, beam components, feeding beam components, operation panel components, dust removal components, and electric box components. The manipulator clamps the panel and sends it to the frame, and then cooperates with the side aligning and the upper and lower drilling components on the beam component to process the panel. The wood chips produced by processing wood fall into the dust removal assembly, and the wood chips accumulated in the dust removal assembly are blown to the dust outlet by the blower, and then sucked away by the dust suction pipe connected to the left side of the frame.

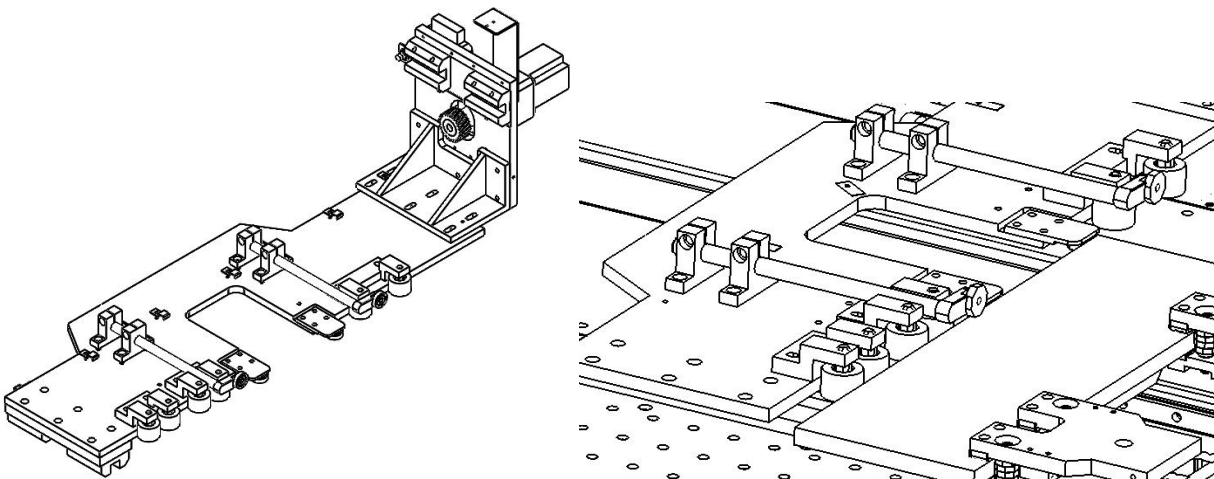
6.2 double clamps assembly



Working principle: Manually push the panel of the air bearing table to make the panel close to the positioning bearing of the manipulator and the positioning cylinder of the frame, as shown in the figure above. Double clamps automatically clamp the plates and feed them into the machine with the upper and lower drilling components and the side supporting components to complete the processing of the plates.

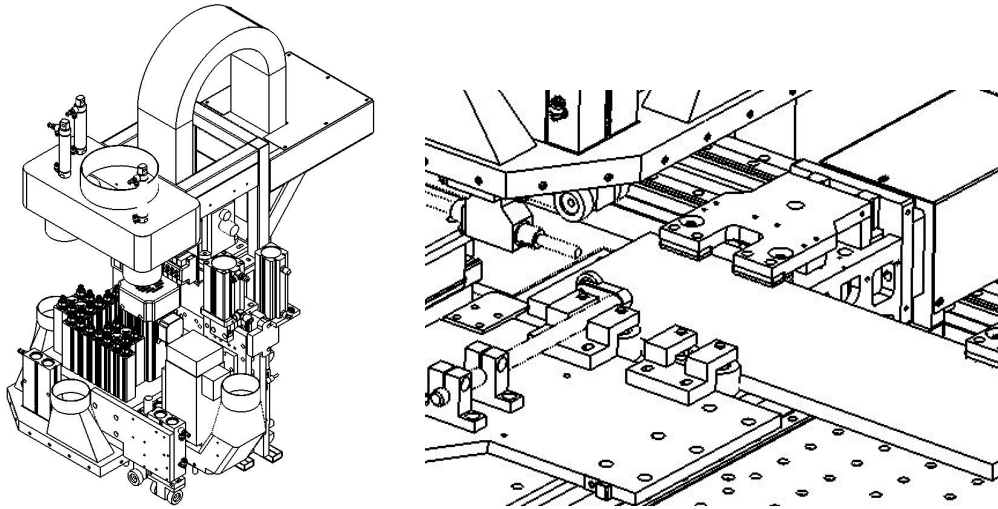


6.3 Side Leaning Components



Working principle: During processing, the side aligning automatically abuts against the side of the plate according to the size of the plate, and the width of the plate can be detected to prevent misplacement of the plate.

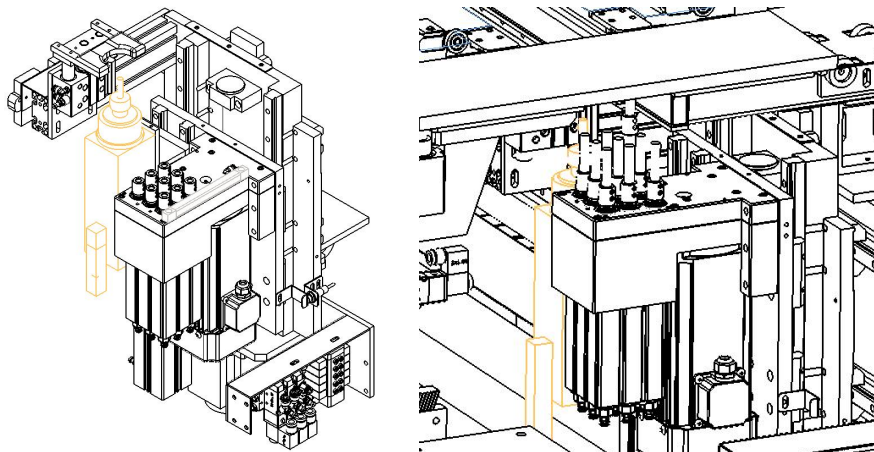
6.4 Up drilling box



working principle:

- (1) The vertical drilling part of the upper drill pack automatically processes the vertical holes on the top surface of the panel, and the horizontal drill automatically processes the four-sided side holes of the panel.
- (2) The spindle of the milling cutter of the upper drilling group automatically grooves, cuts and mills according to the position information.

6.5 Bottom drilling box



working principle:

- (1) The vertical drilling part of the drill package of the lower drilling group automatically processes the vertical holes on the bottom surface of the plate.
- (2) The spindle of the drill-down milling cutter performs grooving and other processing on the bottom surface of the plate according to the position information.

7. Operation Guidance

7.1. Preparation before booting

- (1) Check the equipment operation records, clarify the equipment status, and use it only after the fault is repaired.
- (2) Check whether the lubricating oil of the automatic lubricator of the equipment is sufficient.
- (3) The total air intake of the equipment needs to be kept dry for a long time
- (4) Check whether the drill bit is sharp and whether it is installed firmly. It can be used only after confirmation.
- (5) Check whether the number and specifications of the transferred parts from the previous process are consistent with the order.
- (6) Check whether the air pressure is in the normal range $> 0.7\text{MP}$.

Note: The drill bag needs to be added with grease Klüber L32-CN every 120-150 hours of operation (each nozzle can be injected with a grease gun once, and the specified brand and model grease must be added)

7.2. Processing

- (1) According to the quality and material requirements of the processed products, select the appropriate drill bit.
- (2) After the operator has placed the materials, he should make sure that he and other people are safe before pressing the start button. In case of emergency, press the emergency stop button immediately.
- (3) The small pieces of material in the machine are not allowed to be aligned by hand, and must be pressed and positioned with tools or wooden sticks to avoid side-pressing the machine to injure the hands.
- (4) It is strictly forbidden to process wood with unqualified and waste drill bits.
- (5) When leaving the machine, changing the drill bit, adjusting the machine or before maintenance, the power supply, that is, the main switch of the machine, must be turned off.

7.3. Inspection

- (1) According to the drawings or technical requirements, the first product can be manufactured after self-inspection by the operator and re-inspection by the quality inspector.
- (2) During the processing, the quality inspector will conduct random inspections according to the regulations, and notify the host to stop and adjust when out-of-tolerance and other quality problems are found.

7.4. After finish work

- (1) Turn off the power of the machine and clean up the debris and dust after processing.
- (2) Maintain the equipment, clean up the drill dust and dust inside and outside the machine, and maintain a civilized production environment.
- (3) Fill in the equipment operation records, and notify the successor and dispatch in time if there is a problem.

Untrained personnel are forbidden to operate this equipment. To operate the host, please read the manual before operating.

 Please shut down the computer normally before disconnecting the power. Abnormal power failure will cause system damage

8. Operation instructions

8.1. Boot

- (1) Power on (equipment/vacuum);
- (2) Press the computer start on the operation screen and turn on the industrial computer (when the computer is fully started, proceed to the next step);
- (3) Press the power on the operation screen to start (approximately 1 minute after the next step).

8.2. Software



- (1) Double-click "CncMon32" (enter the host interface machine state: ready-the alarm is not flashing, you can proceed to the next step);
- (2) After the above operations are completed, minimize the current interface.

8.3 Main control interface



- (1) Double-click "HHcnc" (enter the operation interface);
- (2) Click or → select path  or  → select the corresponding file format → import;
- (3) Switch to scan code mode (complete preparation work) Use scan gun to scan code and place the board;
- (4) Matters needing attention: The safety of personnel must be ensured when inserting the board before pressing the start button.

8.4 Finished

- (1) Close HHcnc→CncMon32→Close the PC system;
- (2) Turn off the power and stop;
- (3) Turn off the main power after the PC is completely shut down.

Note: The interval between continuous power on and off must be greater than one minute

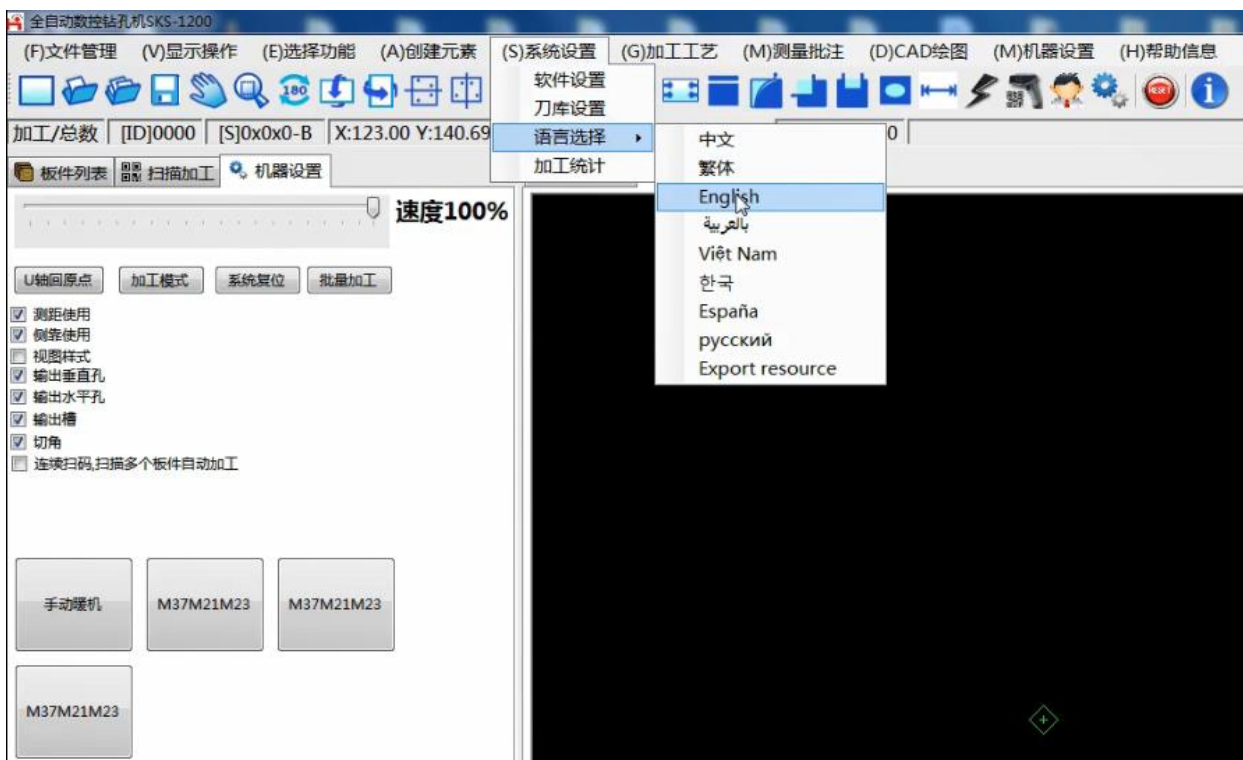


Please shut down the computer normally before disconnecting the power. Abnormal power failure will cause system damage

9. Language switch

9.1. "HHcnc" Operation Guidelines

Double-click "HHcnc"  (enter the operation interface), click (S) in the process area, system settings → language selection → English



9.2. "CncMon32" Operation Guide



(1) Double-click "CncMon32" (enter the operation interface)

(2) Click F8 System Management



(3) Click "F3 parameter setting"



(4) Click "F5 to jump to parameter number"



(5) Enter "3209" in the "Enter parameter number" window, and click "OK"



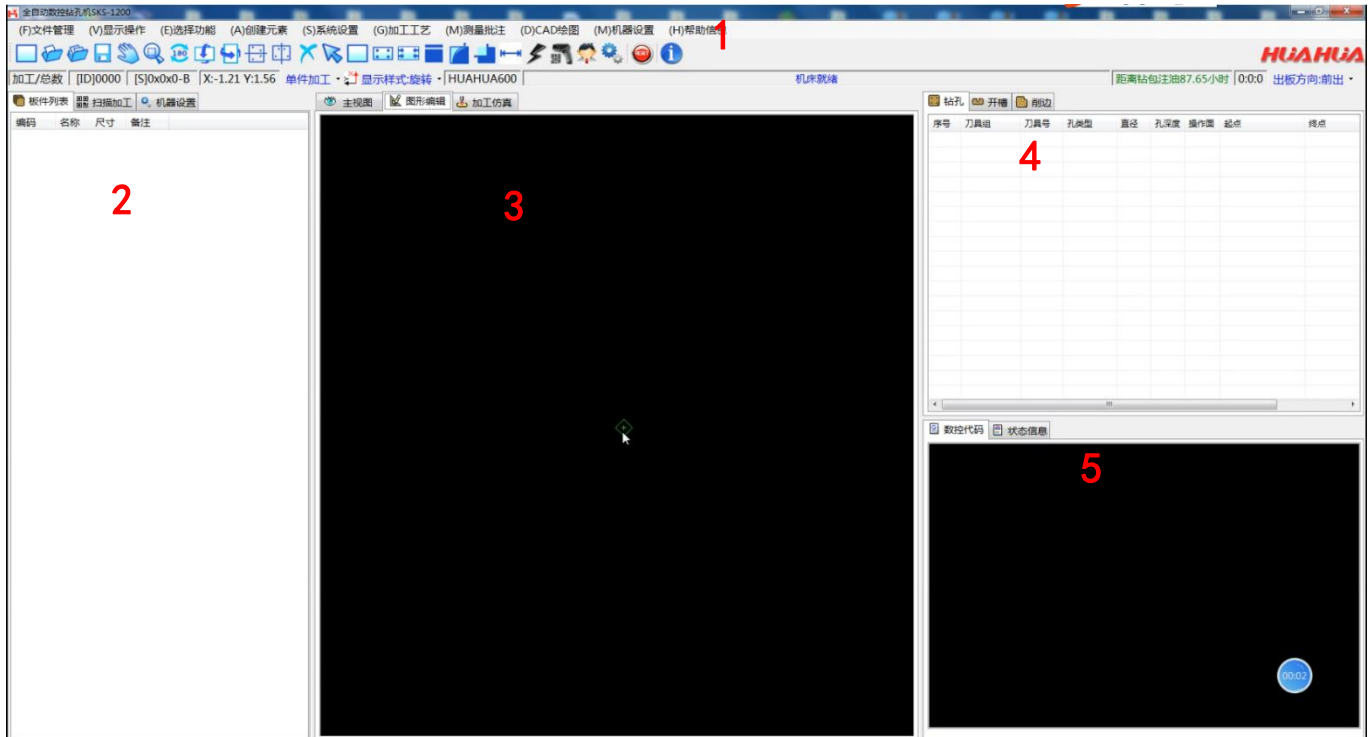
(6) Enter "0" in the numeric field of the number "3209" to convert it into English.



10. Sheet craft creation:

Process creation includes vertical holes, horizontal holes, slots, milling, rectangular milling, circular milling, corner chamfering, Lamino, and Locking. The creation methods are almost the same. Here we take holes, slots and milling as examples.

(1) “HHcnc” home page



1. Process area:

- New create
- Import files/directories
- Save
- Pan/Zoom
- Plate rotation
- Flip and mirror
- Delete/Select
- New board
- Hole/slot/milling creation
- Size
- Generate CNC code
- Scan processing
- Knife library
- Settings
- System reset

2. Panel list/scan processing

3. Plate display:

Main view

Graphic editing

Processing simulation

4. Sheet information

5. Processing status display:

CNC code

Status information

10.1 Create a vertical hole:

Create plate → click vertical hole → select reference point → select process point → set parameters

A: Click on the new board (Ctrl+N):



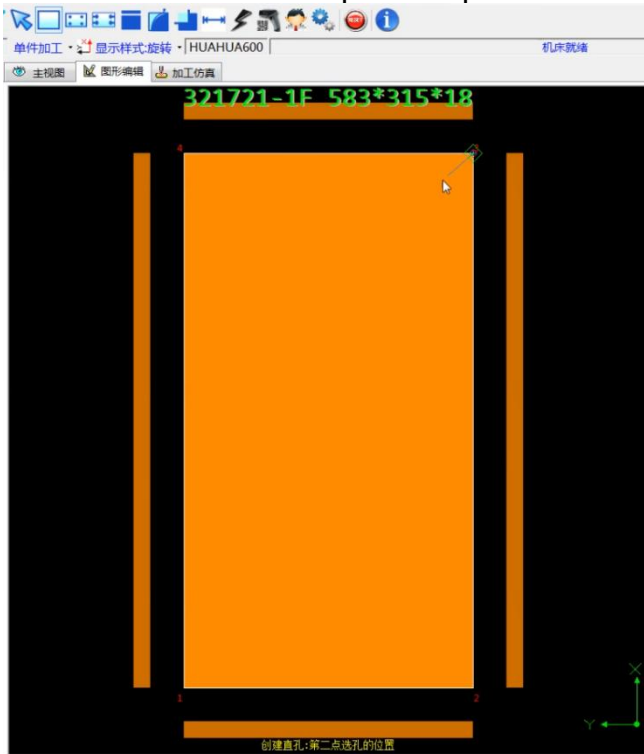
B: Input panel size and number → click "OK":



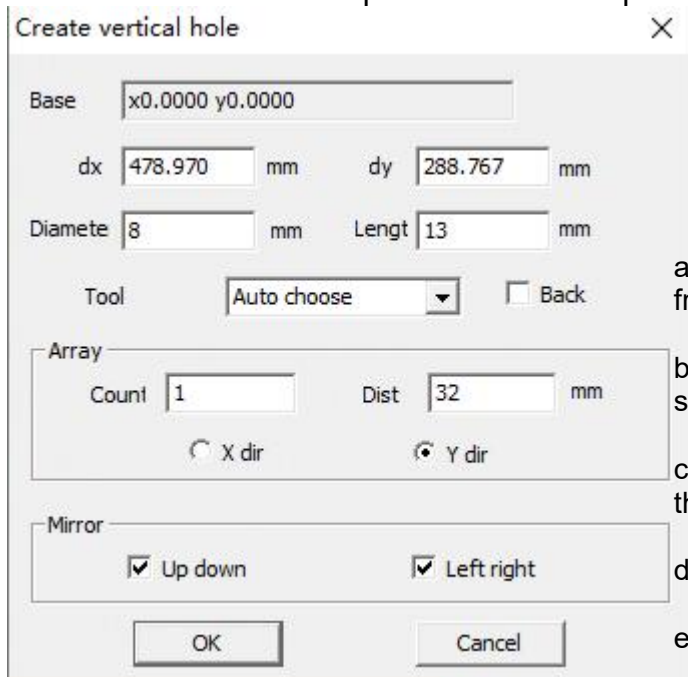
C: Click "Vertical Hole (V)" to create a vertical hole for the plate:



D: Select the reference point on the board (usually select 4 corners of the board), click on the board to select the process point:



E: Set the vertical hole parameters of the plate:



a. Set the coordinates of the hole, (distance from the reference point)

b. Set the diameter and depth of the hole, select the front and back

c. Set the designated tool, if you don't specify the tool, you don't need to set this parameter

d. Set array parameters, select mirror

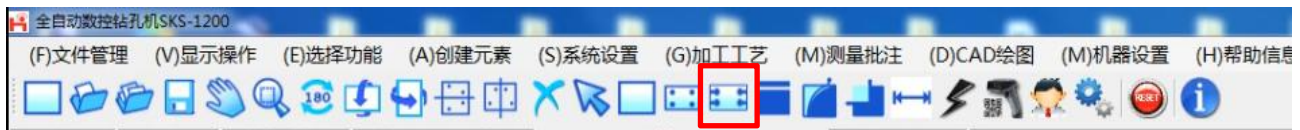
e. Click OK after setting

10.2. Create a side hole (three-in-one hole):

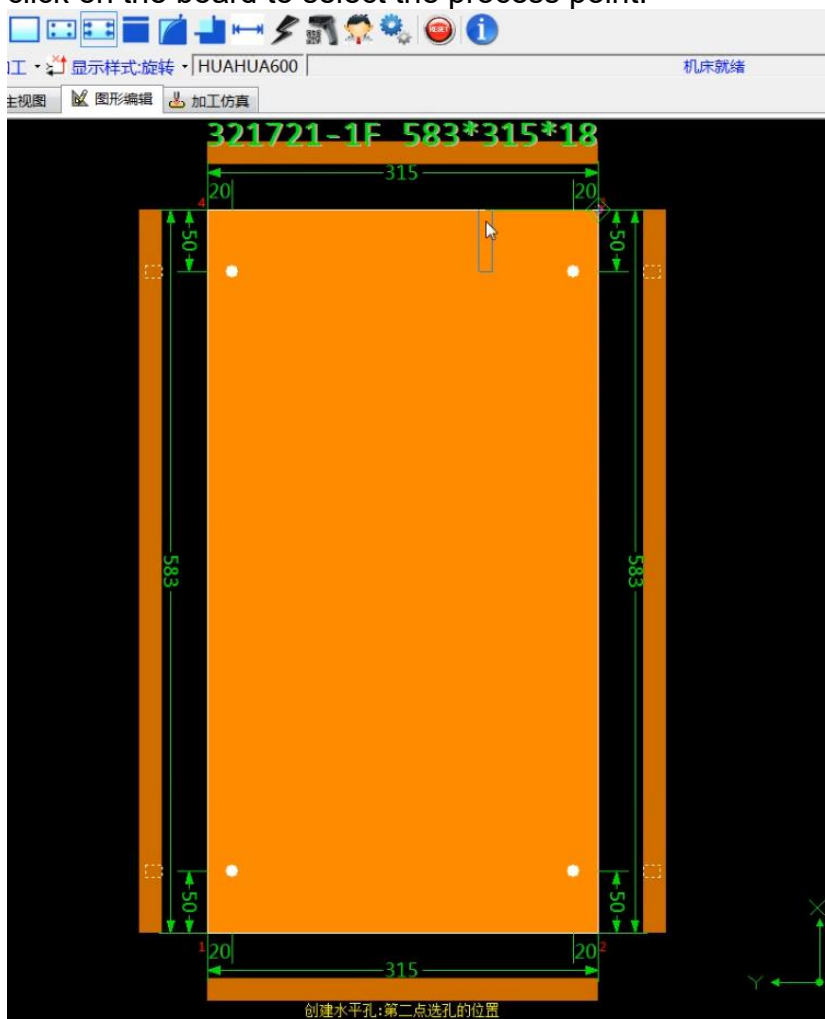
Create plate → click side hole → select reference point → select process point → set parameters

A: To create a plate, please refer to items A and B of "(2) Creating a vertical hole" above for detailed operation.

B: Click "Side Hole (H)" to create a side hole of the plate:



C: Select the reference point on the board (usually select the 4 corners of the board), click on the board to select the process point:



E: Set the side hole parameters of the plate:



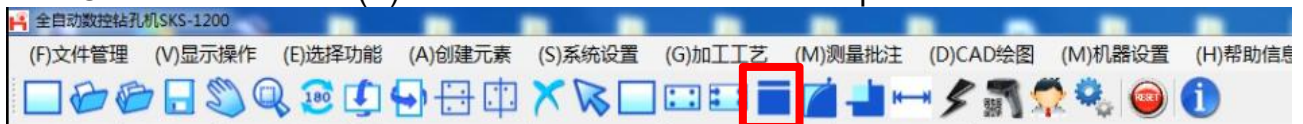
- Set the coordinates of the side hole, (distance from the reference point);
- Set the diameter, depth and number of holes, and spacing;
- Set the designated tool, if you don't specify the tool, you don't need to set this parameter;
- Check "Create Straight Hole" to set the diameter and depth of the straight hole, and specify the straight hole on the front and back;
- Set the array parameters and select mirroring;
- Click OK after setting.

10.3. Create a slot:

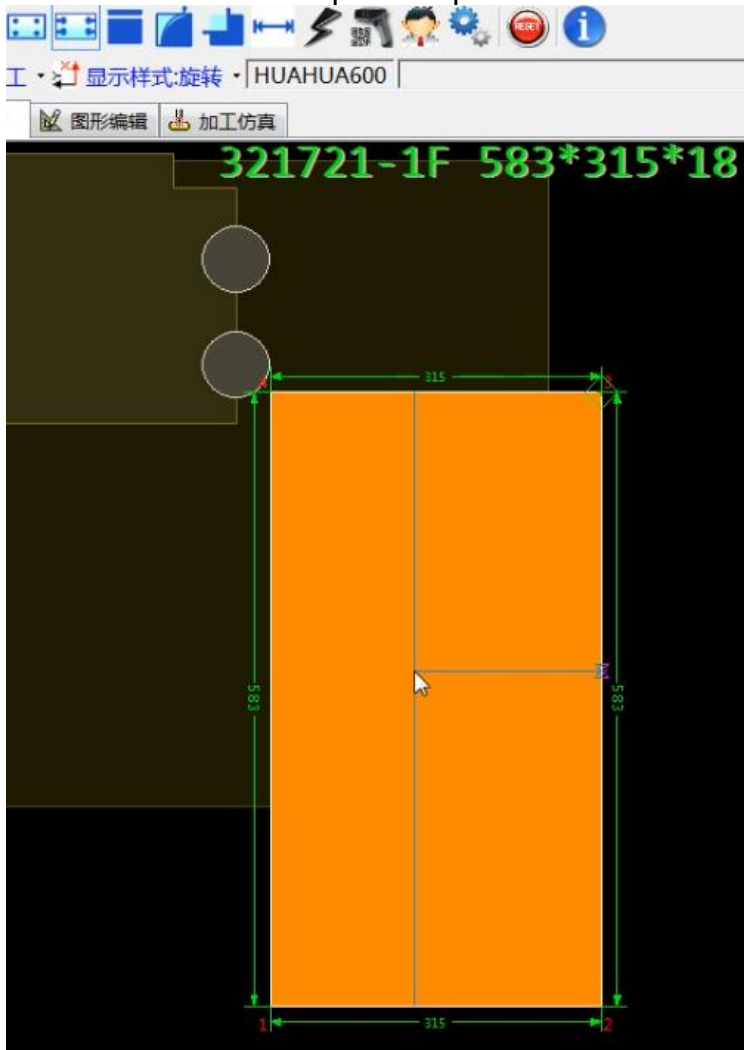
Create plate → click on groove → select reference edge → select process point → set parameters

A: To create a plate, please refer to items A and B of "(2) Creating a vertical hole" above for detailed operation.

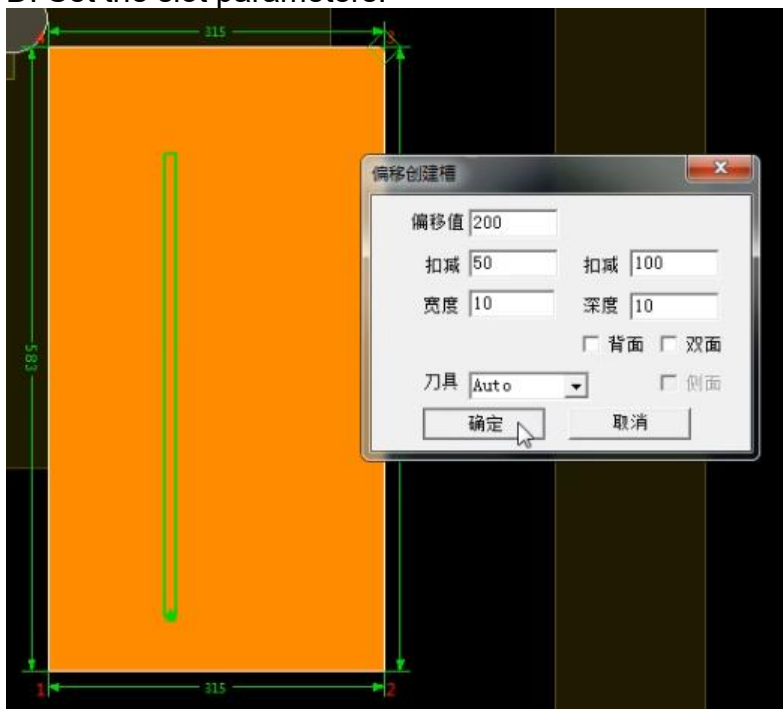
B: Click "Vertical Hole (V)" to create a vertical hole for the plate:



C: Select the reference edge on the board (usually select 4 sides of the board), click on the board to select the process point:



D: Set the slot parameters:



- Set the offset value of the slot, (distance from the reference point);
- Set the width and depth of the groove, and set the deduction value (the reduced distance between the two ends of the groove);
- Choose the front and back
- Set the designated tool;
- Click OK after setting

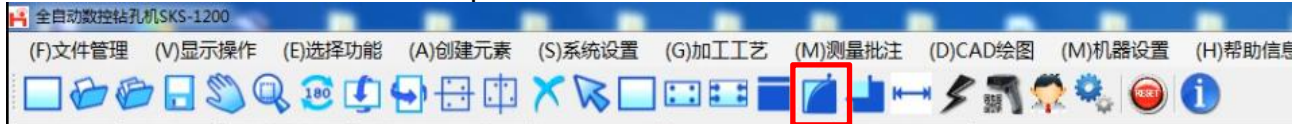
10.4 Create chamfer and chamfer:

The chamfering angle is divided into three types: general oblique missing angle, oblique missing angle, and arc missing angle. The setting method is the same. Here, we take the right angle missing angle creation method as an example.

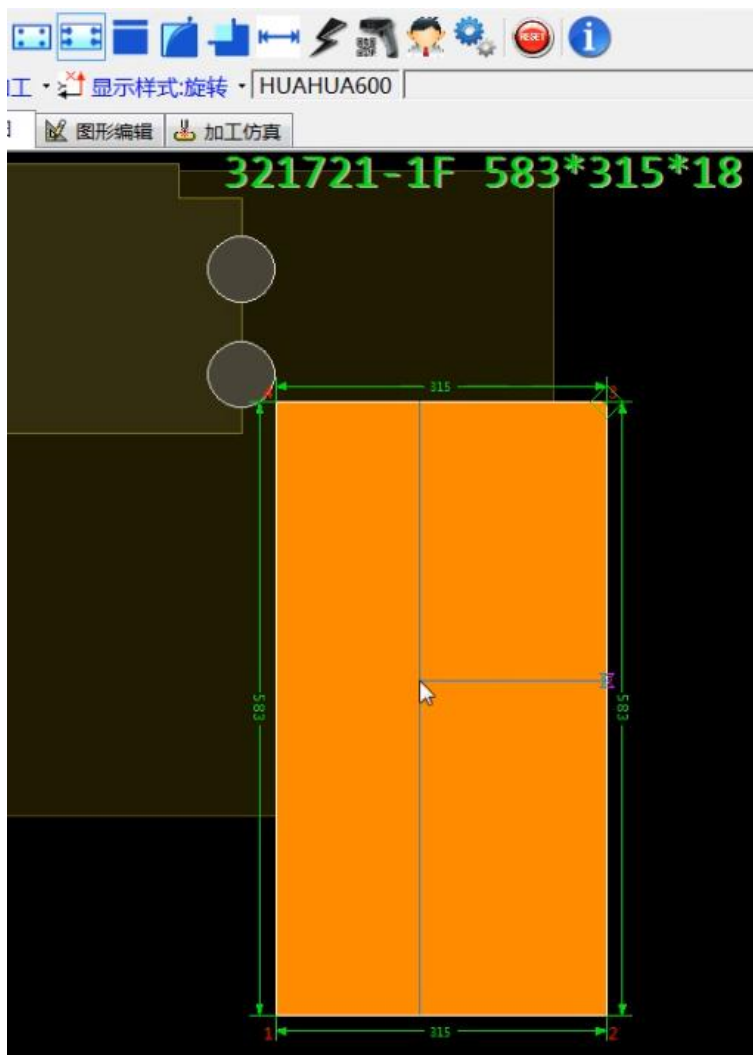
Create plate → click corner chamfer or chamfer → select reference point → select process point → set parameters

A: To create a plate, please refer to items A and B of "(2) Creating a vertical hole" above for detailed operation.

B: Click "Chamfer" to create a plate chamfer:



C: Select the reference point on the board (usually select the 4 corners of the board), click on the board to select the process point:



D: set chamfering parameters

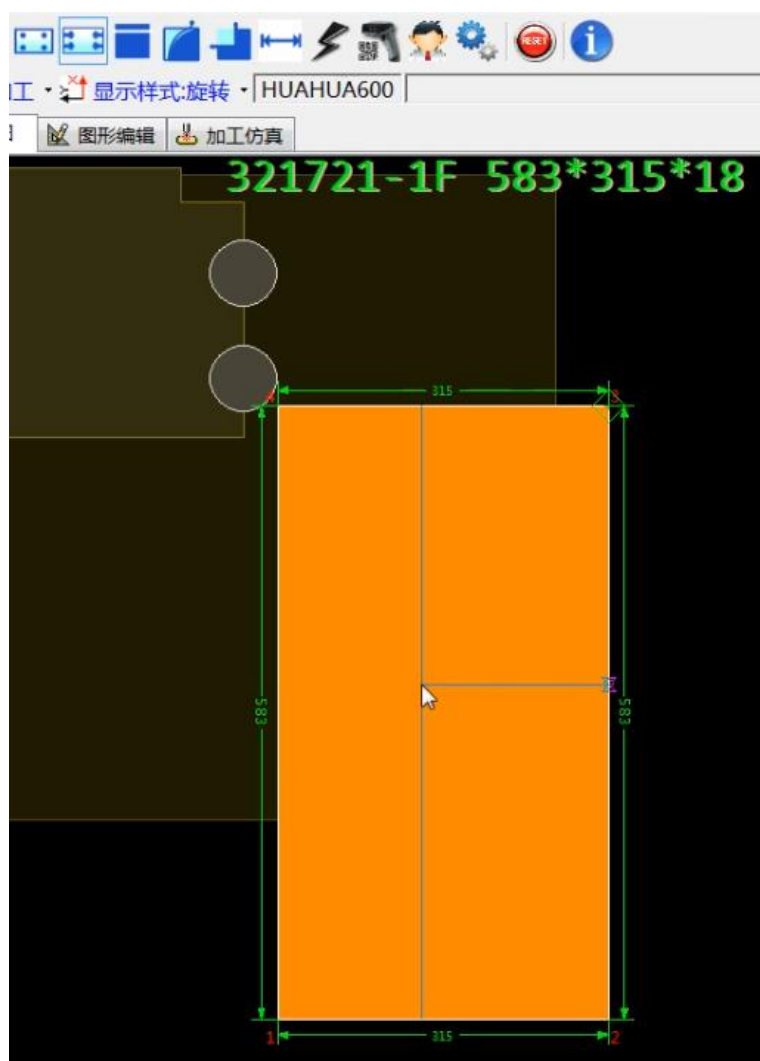


- Set the radius value of the chamfer;
- Select mirror
- Click OK after setting.

E: Click "Chamfer" to create a chamfered corner of the panel:



F: Select the reference point on the board (usually select 4 corners of the board), click on the board to select the process point:



G: Create a chamfer



- Set the value of the chamfer;
- Choose the type of chamfer;
- Select mirror;
- Click OK after setting.

10.5. Create a milling pattern

The milling type is divided into four types: general milling, circular milling, rectangular milling and the above corner cutting. The general milling, circular milling, and rectangular milling have the same settings. Here, the rectangular milling creation method is used as an example.

Create sheet → click "CAD drawing", draw a rectangle → enter coordinate points → enter the rectangle size, select "inverted right angle" → click "select" select rectangle → click "(A) create element" and select "set the selected element as a slot" → Set slot parameter attributes

A: To create a plate, please refer to items A and B of "(2) Creating a vertical hole" above for detailed operation;

B: Click "CAD drawing", click the rectangle;



C: Enter the rectangle coordinates;

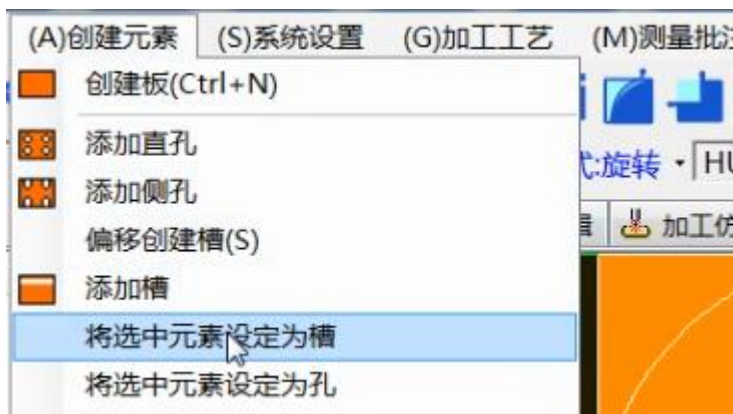


D: Enter the rectangle size:



E: Click "Select" to select the edge of the rectangle

F: Click "Create element", select "Set selected element as slot"



G: Set slot parameter attributes



- a. Set the groove width and depth;
- b. Choose the front and back;
- c. Click OK after setting.

11. Routine maintenance

11.1. Lubrication

11.1.1. Inside the oil tank (46# hydraulic oil), mainly for oil injection

(1) Cylinder-type oil injection system, all s are supplied with independent cylinders to ensure that all s are evenly injected with "46# hydraulic oil" every time



(2) Manual start of rail lubrication



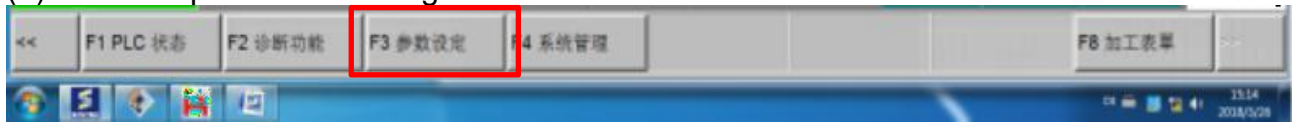
Note: Check the actual amount of oil on the track during use. If the amount of oil added is insufficient, please click the "Lubricant" button under the "M" button interface (valid in any mode). You can add more, but you can't skip it.

11.1.2 Track automatic oil injection system, oil volume adjustment

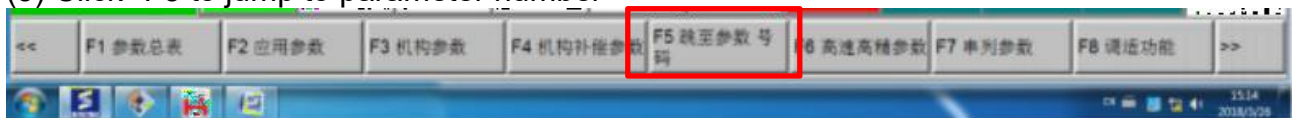
(1) Click "F8 System Management" or ">>"



(2) Click "F3 parameter setting"

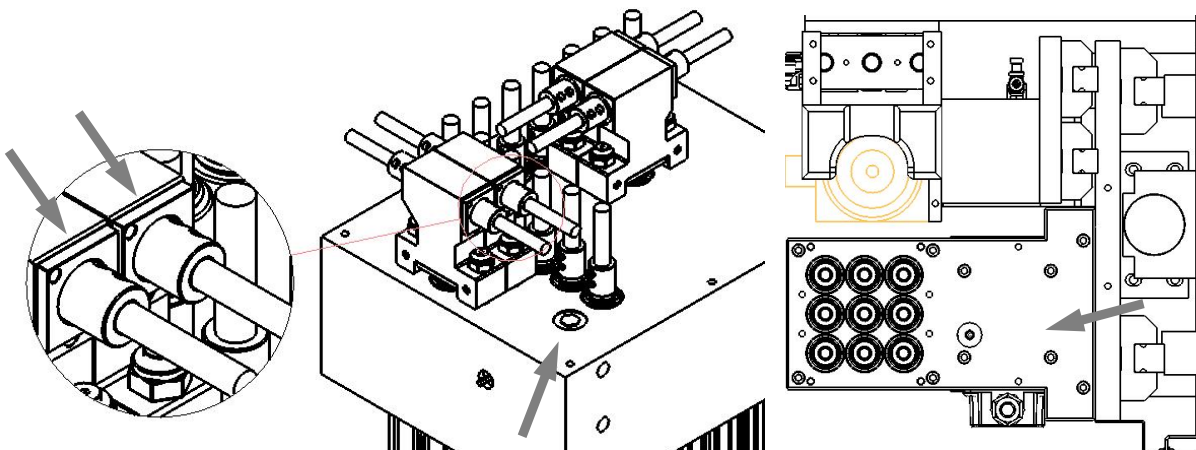


(3) Click "F5 to jump to parameter number"



Jump to No. 3430; set automatic oiling system	time interval	(rail lubrication)
Jump to No. 3431; set up automatic oiling system	Oil filling time	(rail lubrication)

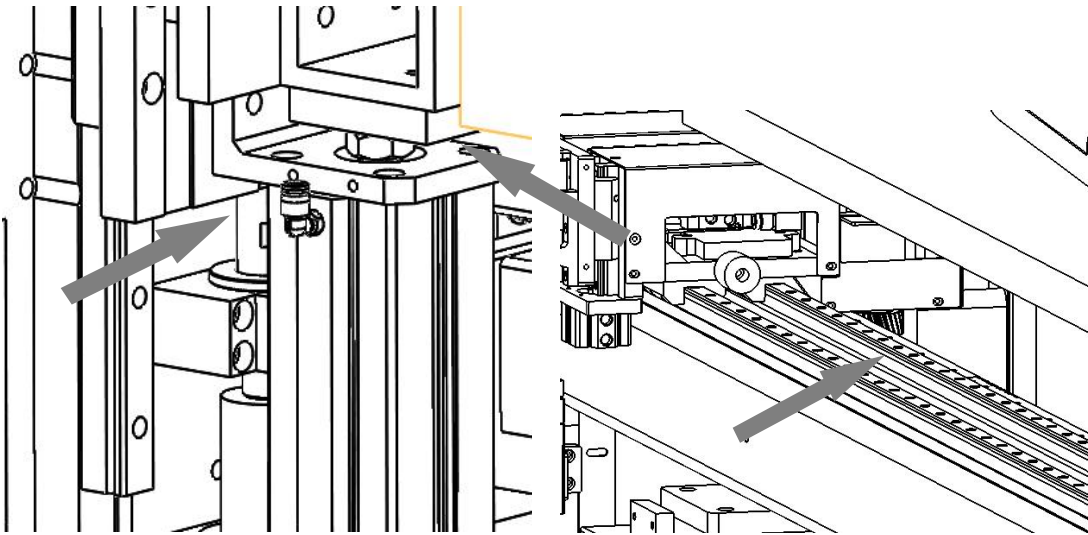
11.1.3 Oil filling of the drill box



- (1) The drill bag needs to add grease Klüber L32-N every 120-150 hours of operation (specified brand and model grease must be added)
- (2) Drilling up: main oil nipple*2, horizontal tool oil nipple*4
- (3) Drilling down: main nozzle *1

11.2 Dust cleaning

The location in the figure is the location that must be cleaned for each maintenance to prevent dust accumulation in long-term use



11.3. Tool change

电

气动元件

固定元件

刀具库

界面按钮

IO状态

用户管理

7

Φ10

8

Φ15

9

Φ8

10

Φ10

11

20+

Φ8

19+

Φ8

20-

Φ8

19-

Φ8

ID	名称	类型	标识	电机	同轴	直径	长度
23	23	1000	0	0		8	
24	26	1000	0	0		5	
25	25	1000	0	0		8	
26	24	1000	0	1		10	
27		0	0	1		0	
28		0	0	1		8	
29		0	0	1		8	
30		0	0	1		8	
31		0	0	1		8	

Note: After the tool replacement is completed, please modify the diameter in the tool magazine to be consistent with the diameter of the actual installed tool

Precautions for tool installation:

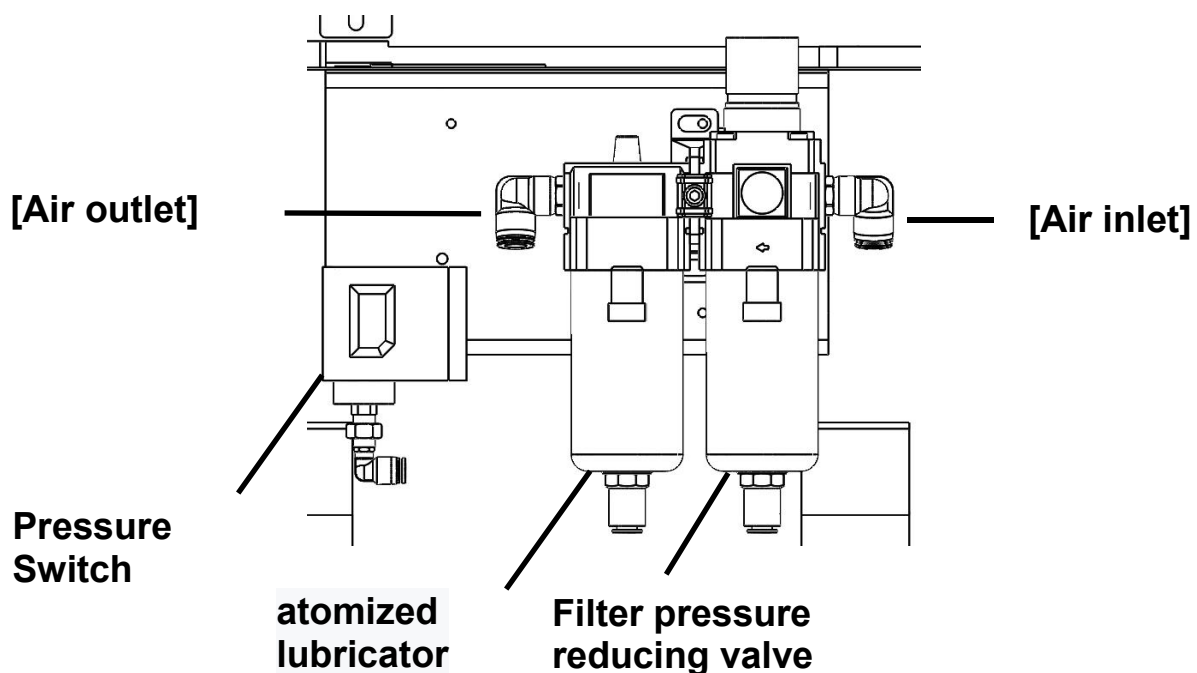
- (1) The installation height should not be too large (no more than 1mm)
- (2) If the installed top wire is slipping, it needs to be replaced immediately
- (3) The installation of the top wire needs to be locked in place to prevent the drill bit from falling off for a long time.
- (4) It is forbidden to use discarded drill bits for installation and processing
- (5) It is forbidden to install a drill with a diameter greater than Φ10mm on the No. 1 tool

Diameter 5 processing speed 4500
Diameter 6 processing speed 4500
Diameter 8 processing speed 4500
Diameter 10 processing speed 1-4000
Diameter 12 processing speed 1-4000
Diameter 15 processing speed 1-2200
Diameter 20, processing speed 1-1200

Milling cutter: The installed protruding length of the milling cutter is greater than 45, which is ideally close to the length before removal

Depth adjustment: the milling cutter in the tool magazine corresponds to the value of "+Z", the larger the value, the deeper the groove, the upper and lower milling cutters are common

11.4 Pneumatic equipment inspection



- (1) Check whether the air inlet/outlet connection pipeline of the equipment is leaking;
- (2) Check the liquid level in the filter tank, if it exceeds the highest scale line, pour out the filled liquid;
- (3) Check the liquid level of the triple lubricator, if it is lower than the lowest scale line, add pneumatic oil to the highest scale line;
- (4) Daily inspection and drainage treatment for the gas storage tank;

11.5. Daily maintenance

- (1) Check whether the lubricating oil of the automatic oil feeder of the equipment is sufficient, and check the actual oil filling amount on the track;
- (2) Check the oil filling time between the machine and the last drilling bag to ensure that it can be maintained normally during each period of time;
- (3) Turn off the power of the machine, and clean up the debris and dust after processing in time; (the wooden block must be cleaned after the milling is completed)
- (4) Maintain the equipment, clean up the drill dust and dust inside and outside the machine, and maintain a civilized production environment;

Fill in the equipment operation records, notify the successor and dispatch in time if there is a problem

12. M code function

Item	name	M code			OUTPUT
1	Work indicator		R65.0	Y0	Y0.0
2	Pause indicator		R65.1	Y1	Y0.1
3	Upper spindle start	M3/M5	R65.2	Y2	Y0.2
4	Start up the drill package	M6/M7	R65.3	Y3	Y0.3
5	Lower spindle start	M13/M15	R65.4	Y4	Y0.4
6	Drill down package start	M16/M17	R65.5	Y5	Y0.5
7	Fan start	M26/M27	R65.6	Y6	Y0.6
8	Oil pump		R65.7	Y7	Y0.7
9	Positioning cylinder	M36/M37	R65.8	Y8	Y0.8
10	Left hand 1	M20/M21	R65.9	Y9	Y0.9
11	Left hand 2		R65.10	Y10	Y0.A
12	Right hand 1	M22/M23	R65.11	Y11	Y0.B
13	Right hand 1		R65.12	Y12	Y0.C
14	Milling cutter pallet	M64/M65	R65.13	Y13	Y0.D
15	Upper spindle cylinder solenoid valve	M40/M41	R65.14	Y14	Y0.E
16	Lower spindle cylinder solenoid valve	M60/M61	R65.15	Y15	Y0.F
17	Pressure wheel 1	M28/M29	R66.0	Y16	Y1.0
18	Pressure wheel 2	M50/M51	R66.1	Y17	Y1.1
19	Pressure wheel 3	M32/M33	R66.2	Y18	Y1.2
20	Pressure wheel 4	M34/M35	R66.3	Y19	Y1.3

13. Signal detection I/O point

Item	name	M52 detection point	alarm code			INPUT
1	Program start			R60.0	X0	X0.0
2	time out			R60.1	X1	X0.1
3	reset button			R60.2	X2	X0.2
4	Tool change key			R60.3	X3	X0.3
5	Emergency stop		R44.0	R60.4	X4	X0.4
6	Foot switch			R60.5	X5	X0.5
7	Air pressure switch		R44.3	R60.6	X6	X0.6
8	X limit (X1)			R60.7	X7	X0.7
9	U limit (X2)			R60.8	X8	X0.8
10	Y limit (Y1)			R60.9	X9	X0.9
11	V limit (Y2)			R60.10	X10	X0.A
12	Z limit (Z1)			R60.11	X11	X0.B
13	W limit (Z2)			R60.12	X12	X0.C
14	A limit (side lean)			R60.13	X13	X0.D
15	X1 hand signal	I0		R60.14	X14	X0.E
16	X2 hand signal	I1		R60.15	X15	X0.F
17	Upper cutter upper	I2		R61.0	X16	X1.0
18	Down milling cutter	I3		R61.1	X17	X1.1
19	Pressure roller 1 signal	I4		R61.2	X18	X1.2
20	Pressure roller 2 signal	I5		R61.3	X19	X1.3
21	Pressure wheel 3	I6		R61.4	X20	X1.4
22	Press wheel 4 signal	I7		R61.5	X21	X1.5
23	Level 1 signal	I8		R61.6	X22	X1.6
24	Level 2 signal	I9		R61.7	X23	X1.7
25	Level 3 signal	I10		R61.8	X24	X1.8
26	Level 4 signal	I11		R61.9	X25	X1.9
27	Fan overload			R61.10	X26	X1.A
28	Overload of drill bag			R61.11	X27	X1.B
29	Drill down package			R61.12	X28	X1.C
30	Upper spindle inverter		R44.9	R61.13	X29	X1.D
31	Upper spindle		R44.12	R61.14	X30	X1.E
32	Lower spindle inverter		R44.10	R61.15	X31	X1.F
33	Lower spindle		R44.13	R62.0	X32	X2.0
34	The upper drill box		R44.6	R62.1	X33	X2.1
35	The drill box motor		R44.7	R62.2	X34	X2.2
36	Phase sequence			R62.3	X35	X2.3
37	Fiber inspection	I12		R62.4	X36	X2.4
38	Positioning cylinder	I13		R62.5	X37	X2.5
39	Overheating of the			R62.6	X38	X2.6
40	Overheating of the			R62.7	X39	X2.7

14. Operate key combinations

Ctrl+2 automatic

Ctrl+4 Continuous manual (In this mode, all functions of the "M" button can be used)

Ctrl+7 origin (Each axis can be operated to move to the mechanical origin position)

Ctrl+Alt Reset/retract/bounce alarm screen

15. Operation definition description



(1) When a pause occurs during processing, open "CncMon32" and click to enter processing monitoring



(2) When a certain point of failure is detected, the machine tool stays at the current position (as shown in the figure below), and the yellow part is the part that needs to be checked

Note: 71570 (line number) M52 (general detection code) I31 (signal point No. 31) P1 (signal)/P0 (no signal)

```

04P1000
71530 M52 I29 P1 //17号刀具到位检测信号灯亮起
04P1000
71560 M52 I30 P1 //18号刀具到位检测信号灯亮起
04P1000
71570 M52 I31 P1 //19号刀具到位检测信号灯亮起

04P1000
72310 M52 I16 P0 //检测暂停按钮 (I点不通状态)
04P1000
72310 M52 I17 P0 //检测启动按钮 (I点不通状态)

```

15.1.1. Independent tool retrieval steps

- (1) Machine status: ready, no alarm
- (2) Open the CNC interface → "M" button
- (3) Select "Continuous Manual" mode
- (4) Click on the functional part you need to call
- (5) After the operation is completed, you must use Ctrl+Alt to reset to ensure that each component is in the reset state

15.1.2 MDI input use (example)

Example: (You can directly input the following code into MDI for use)

```

G00 A0.00 //Quickly move A axis to 0mm position
T1 //No. 1 tool is raised
T1T2T3 //No.1, No.2, No.3 tool is raised
G04P1000 //Delay for 1 second
T00 //Retract all tools
M50 //The front pressure wheel is pressed down
G04P500 //Delay 0.5 seconds
M51 //The front pressure wheel is raised
M27 //End code

```

15.1.3 Servo motor code movement command editing

Quick move command

- (1) Example G00 X100.00 to move the X axis to the position specified speed movement command of 100MM quickly
- (2) Example G01 X100.00F5000 moves the X axis to the position of 100MM at a speed of 5 meters
- (3) Example: (The following code can be directly input into MDI for use)
G00A0.00 //Quickly move A axis to 0mm position

M00 //Single block stop means stop in this line, if you need to execute the next step, please press the green start button once
 G01Y900.00F15000 // Move Y axis to 900mm position at 15m speed
 M00 //Single block stop means stop in this line, if you need to execute the next step, please press the green start button once
 G00X500.00 //Quickly move the X axis to the position of 500mm
 M27 //End code

Note: Please ensure that every trip is operated in a safe state

15.1.4 Signal diagnosis view

Open the CNC interface → click F8 system management → F2 diagnosis function → F5 input diagnosis

15.1.5 The emulator cannot be opened

(1) If the simulator CncMon32 is turned on, the following conditions will appear:



(2). Click F6 to end or check the F2 IP connection. It is forbidden to press other buttons. Check whether the background system is fully turned on. After the background system is fully turned on, start the PC emulator

15.1.6 The PC emulator is not connected to the system background

If the PC emulator is not connected to the system background, please perform the following actions to reconnect to the background:

(1) F5 maintenance



(2)F2 network settings



(3) F4 back connection setting



(4) F2 IP connection

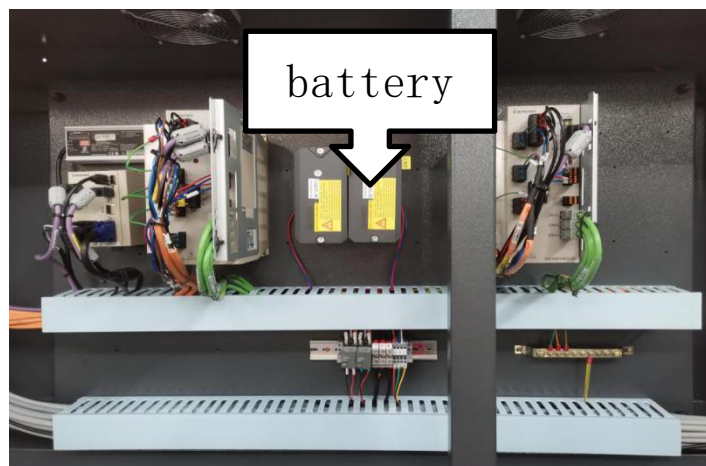


(5) Restart the emulator after it prompts that the storage is successful; if it prompts that the connection fails, please check whether the system is turned on normally and whether the network cable connection is normal.

15.1.7 Replacement battery

Drive and control integrated system battery replacement

(1.2|3.0|3.1|3.2|3.3|6.0|6.1|6.3|6.5 version)



The service cycle is 3-6 months, and the battery in the battery box needs to be replaced (3 AA dry batteries 1.5V)

Precautions:

(1) Alarm "please contact after-sales service, need to replace the battery" processing method

- (2) When opening the battery cover, please take a photo and record the positive and negative direction of the battery.
- (3) Nanfu No. 1 battery must be used.

- (1) Press Ctrl+7. Switch to "Back to Origin" mode;
- (2) Press the emergency stop button;
- (3) Long press Ctrl+Alt (about 6 seconds) until the alarm is removed;
- (4) Normal use and replace the battery in the power-on state.

16. Common faults and solutions

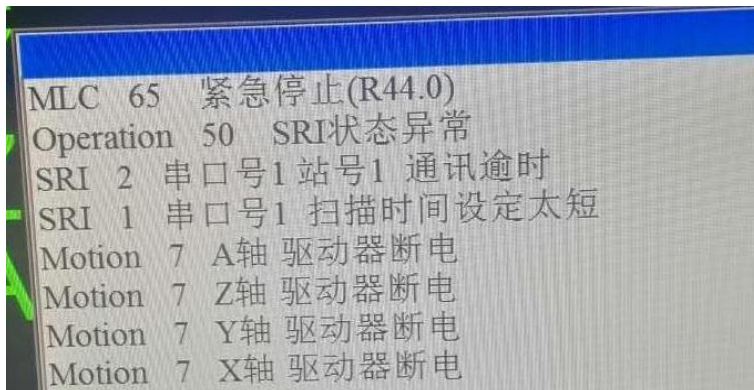
16.1 Failure shutdown

The machine stops due to failure during processing, please confirm the situation of the problem first:

- (1) The hardware has collision, please reset the machine first, and check whether the collision part is complete or out of position;
- (2) There is no hardware collision, please confirm the current displayed alarm to solve;
- (3) Confirm that the state of the machine tool is "Ready". If it is "Not Ready", please restart the lower machine position system after confirming the current machine alarm;
- (4) Insufficient air pressure;
- (5) Exceeding the positive|negative travel limit;
- (6) Missing position command | Excessive torque | Yes? The axis is not enabled.

16.2 Incomplete shutdown

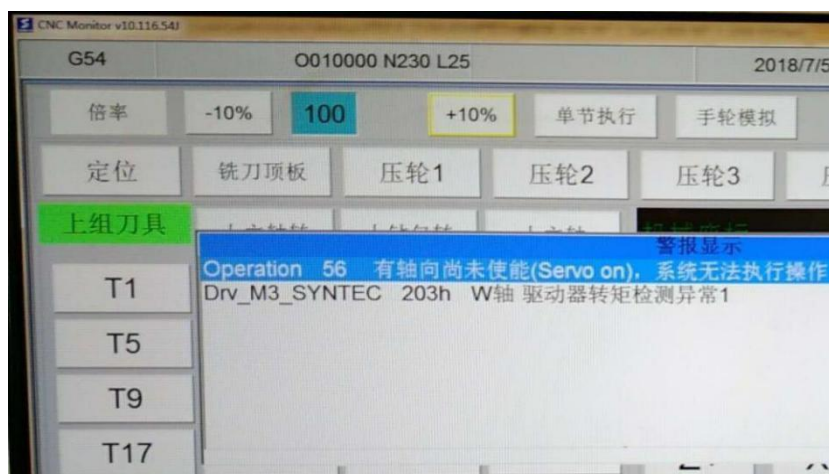
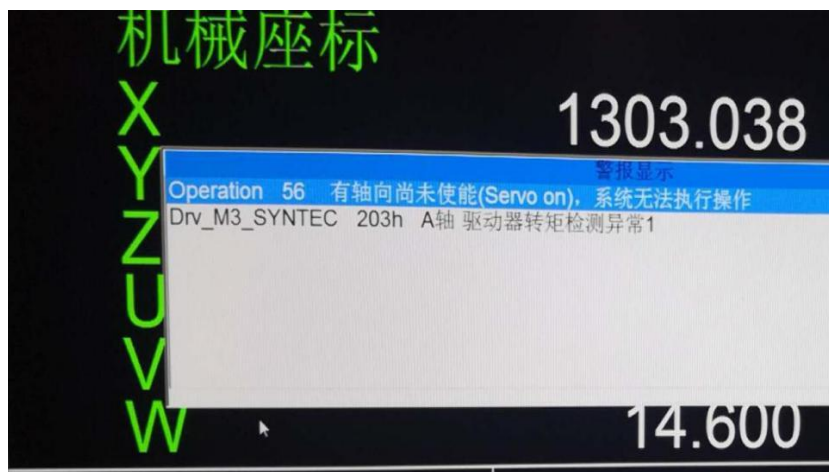
The solution to the situation (as shown below)



solution:

- (1) Close all software (you do not need to close the PC);
- (2) Turn off the key switch (wait 1 minute);
- (3) Turn on the key switch (wait one minute) and open the software normally.

16.3 Alarm "Missing Position Command" or "Axial Disabled" Fault Solution

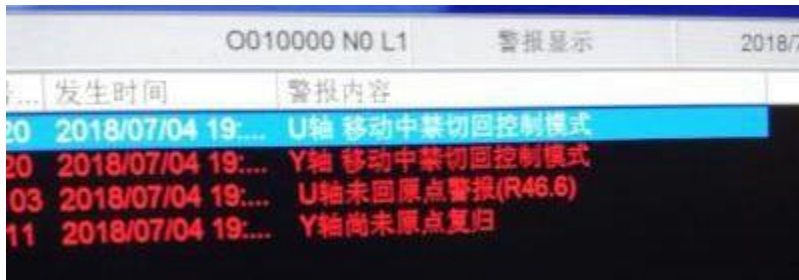


Solution

(1) When the instantaneous force is less than 60% of the motor itself, a "missing position command" alarm will appear, which can be released by resetting (all axes);

(2) When the momentary force exceeds 60%, the motor will be directly independent, the system will no longer be enabled, and it needs to be eliminated through normal restart (all axes).

The situation after pressing the emergency stop button during processing (as shown in the figure below):



16.3.1 Operating status

When the machine is moving in the axial direction during processing, pressing the emergency stop will cause the alarm as shown in the figure above

Approach

- (1) Take out the board;
- (2) Check all hardware parts of the machine and confirm that all parts are free from collision;
- (3) The machine can be automatically restored after normal restart, and it can be used normally.

16.4 Troubleshooting



When the processing status is as follows:

- 1) NC transfer to the lower computer, waiting for the lower computer to process...
- 2) No activation or malfunction

(1). Please open "CncMon32" as shown in the figure below, and click F5 to maintain



(1)

(2). Click F1 alarm display



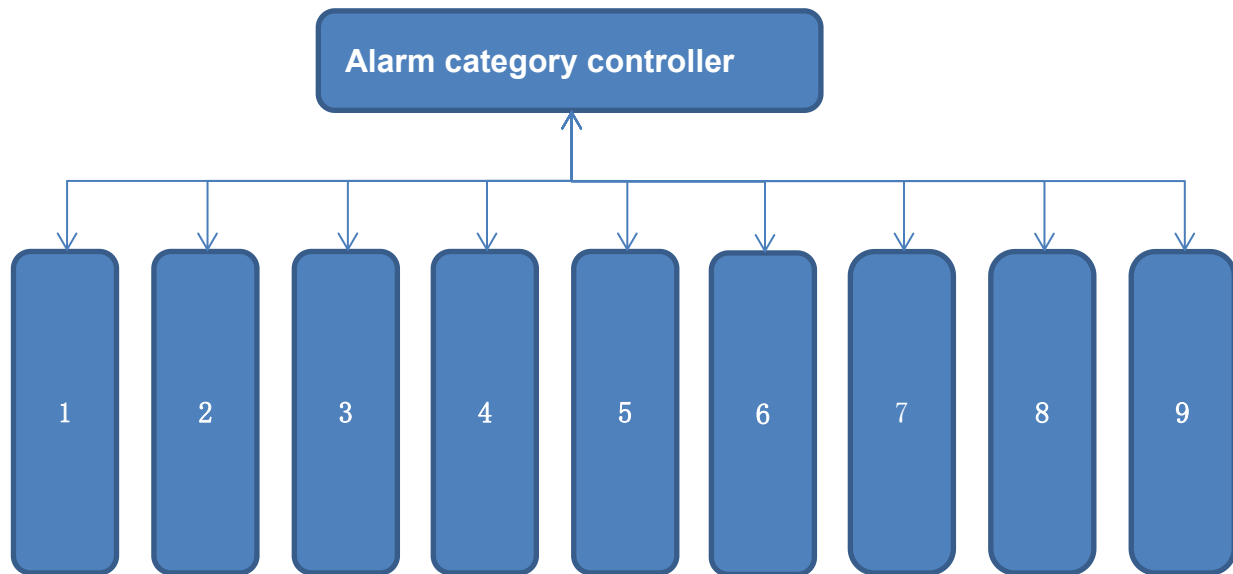
(3). Click F2 historical alarm



(4) Check the alarm record and eliminate the corresponding fault

16.5 Troubleshooting common alarm

16.5.1



1. System Operation Alarm OP
2. Servo axis alarm MOT
3. Syntax check alert COMO
4. Program alarm COR
5. PLC alarm MLC
6. Spindle alarm SPD
7. Macro alarm MAR
8. Turret Alarm ROT
9. SRI alert SRI

16.5.2 OP-018 port parameter setting error

(1) Generation principle: wrong setting of axis card port number for axial (21-40) or spindle (1621-1630); wrong setting of serial PLC axis and ROT port number; parameter 3261-3263 (driver bureau corresponding to M3 IO) No.) Setting error

(2) Possible causes and solutions: The numbers set by parameters 21-40 are not supported or are duplicated (only numbers 17/18 can be set virtually repeatedly)

(3) Check whether Pr3261-3263, Pr21-40, Pr1621-1630 conflict with each other

(4) When using SRI DA spindle (Pr1791-1800 setting 7), the register of Pr1621-1630 is set as non-user-defined area

(5) The time Pr3203 is too small, it is recommended to be set to 3000

(6). Pr13 *The number of axis version is set incorrectly, modify it after inspection

16.5.3 OP-023 Power cut during processing, please re-check the processing data

(1) Generation principle: When the controller cycle starts, the processing flag will be stored back to Fram or REGISTRY.DAT, and returned during processing

(2) Clear when ready, this alarm is issued when the flag is not cleared when the controller is turned on

- (3) Possible causes and solutions: there is a power failure during processing, restart the controller and driver (twice) after the power failure, wait for 30 seconds and then power on again
- (4) The power is not cut off during processing, which may be due to software upgrade

16.5.4 OP-052 M3-IO machine status is abnormal

- (1) The M3-IO card is not connected firmly, check the installation is firm
- (2) Parameter 3261 is set incorrectly and set correctly
- (3) Drive and control integrated to FC parameter settings (using the backup of the drive and control integrated to directly restore to the FC system will alarm OP-52)

Drive and control one	FC system
Pr1621	100601 19
Pr2021	5031 20
Pr3261	1006 0

16.5.5 MOT-011 Drive communication abnormal

Possible causes and solutions:

- (1) Check whether the drive parameter dip switch and the controller parameter (Pr21~) correspond correctly
- (2) The communication lines between the controller and the drive are faulty or loose, replace or install a fixing frame
- (3) No grounding leads to noise, check whether the machine is properly grounded

16.5.6 MOT-008 Missing position command

- (1) After the controller stops outputting a certain axial command for 1 second, check that the error between the feedback command and the output command is not within the predetermined error range, then this alarm will be sent (the settings of Pr561~Pr580 are the missing position commands Inspection scope)

(2) The command transmission line from the controller to the driver, and the feedback line from the encoder to the controller is poorly connected or disconnected (check that the alarm does not shut down after the alarm occurs, first check whether the value of the diagnostic function 8, 9, 10 data has converged to 0, if the value does not change continuously, please search the origin again (no need to restart), after searching the origin, if any of the values in No. 24 25 26 is not zero, there is a problem with the feedback loop; if the diagnosis variable is 40 41 If any value in 42 is not zero, it means that there is a problem with the circuit from the controller to the motor.

- (3) Amplification parameter Pr561

16.5.7 MOT-019 following error exceeded; MOT-023 serious following error exceeded

- (1) Due to the characteristics of the servo, the positioning of the servo motor cannot immediately respond to the commands of the controller, and there will be a backward phenomenon. When the backward amount is not within the allowable range, the controller will send this alarm (allowable amount $\{\max[(\text{each The first stage speed of axis homing}), (\text{G00 speed of each axis})]/Kp\}$)
- (2) The movement of the mechanism is obstructed and the movement is not smooth (lubrication, whether there is foreign matter, whether it is stuck)
- (3) The parameter encoder resolution or electronic gear ratio setting is wrong (check whether the resolution and gear ratio are correct)
- (4) The acceleration and deceleration time of the controller parameter setting is too short (increase the synthetic or axial acceleration and deceleration time)

16.6 ROT alarm

16.6.1 ROT positioning deviation is too large

Possible Causes:

- (1) After ROT starts positioning, the positioning cannot be completed due to the triggering of an emergency stop, or the controller is shut down and restarted, and the system detects that the current position deviation is too large (the difference between the "feedback angle" and the "positioning angle" exceeds the "position check window" Angle" setting value, the default is 0.1 degree).

elimination method:

- (1) Complete the chaotic tool elimination program (manual forward/reverse) to cancel the alarm and repositioning.
- (2) Magnify the position to check the angle setting value of the window

16.6.2 ROT-017 Custom Tool Position Setting Error

Possible Causes:

- (1) In the tool position custom table, the position of the active tool is not set from small to large.

elimination method:

- (1) Check the tool position custom table, whether the tool position is set correctly from small to large.
- (2) Check whether the parameter total number of tools matches the setting of the custom tool position to be enabled.
- (3) Reset the ROT or modify the parameter settings to clear the alarm.

16.7 SRI alert

16.7.1 SRI-004 The module station number or sequence does not match the parameters

Possible Causes:

- (1) The station number or module sequence of the external machine has been changed.

elimination method:

- (1) Reset the system SRI parameters (SRI parameter setting—com—edit—automatic synchronization—save)

16.7.2 SRI-007 Module communication is abnormal

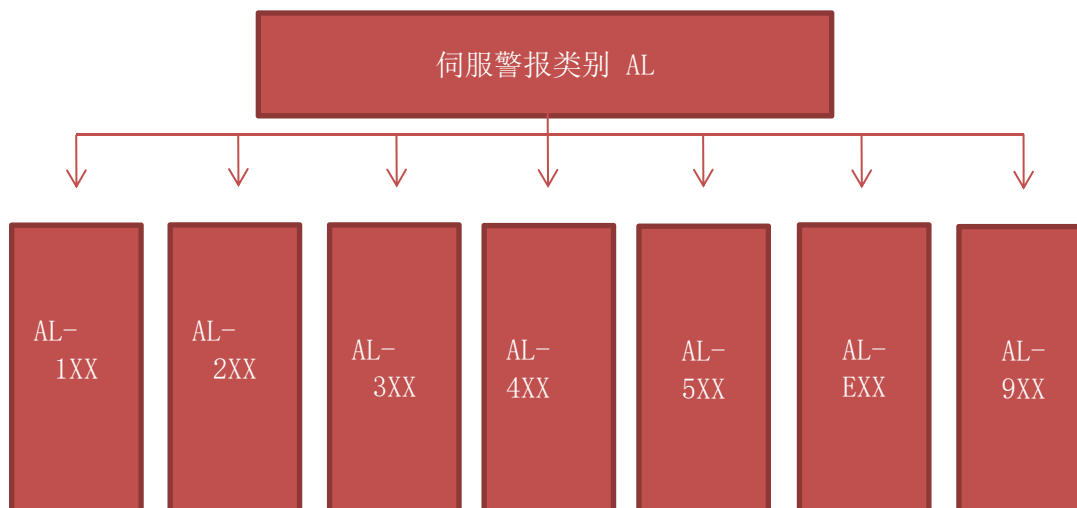
Possible Causes:

- (1) The module cannot communicate normally

elimination method:

- (1) Restart after power failure.
- (2) Update the firmware version.
- (3) If the alarm is skipped for a period of time, it means that it may be noise interference. Please check whether there are interference sources around, and the equipment and wires are anti-interference.
- (4) Whether the treatment is correct.

16.7.3



1. Drive alarm AL-1XX
2. Motor alarm AL-2XX
3. Encoder alarm AL-3XX
4. Tuning alarm AL-4XX
5. Application alert AL-5XX
6. Special alarm AL-EXX
7. Drive warning AL-9XX

16.7.4 AL230 magnetic pole angle deviation is too large

(1) Possible cause: the phase sequence of the power line is wrong

Countermeasures: UVW of the power line corresponds to the UVW of the drive terminal

(2) Possible cause: the encoder polarity setting is wrong, check the encoder polarity

Solution: Set the Pn-021 encoder polarity to 0 when the power line wiring is correct

(3) Possible reasons: the inspection window is too small, check the parameter settings

Countermeasures: enlarge Pn-502 zero speed arrival inspection window

16.7.5 AL-203 drive torque detection abnormal

(1) Possible cause: The UVW phase sequence of the motor power line is wrong

Countermeasures: Correct the phase sequence of the power line or modify the parameters Pn-021 (0 to 1 or 1 to 0)

(2) Possible cause: excessive load during operation due to mechanical factors

Solution: Observe whether the load factor exceeds the over-torque level set by Pn-441 during the alarm

(3) Possible cause: the magnetic pole angle of the motor is wrong

Countermeasures: redo magnetic pole angle detection

(4) Possible cause: motor parameter error

Solution: Check whether the PN7 parameters are consistent with the motor nameplate, and re-import the motor standard parameters

16.7.6 AL-101 drive overload

Possible Causes:

(1) Mechanical factors cause the motor to lock up, resulting in excessive load during operation

(2) Excessive load

(3) Encoder and motor wiring error

(4) Encoder is abnormal

elimination method:

(1) Exclude mechanical factors.

(2) Check whether the diagnostic variable Pn-D30 (D1-16) exceeds the parameter Pn-650 (P5-01) for a period of time, and reduce the load if there is any.

(3) Correct wiring

16.7.7 AL-110 drive severe overvoltage

Possible Causes:

(1) When the motor is decelerating, the braking resistor cannot consume the energy of recovery

- (2) AC power input voltage is too high
- (3) Drive hardware failure

elimination method:

- (1) Check the brake resistance specifications
- (2) Check whether the AC power supply meets the drive specifications
- (3) Excluding the above two, there may be a hardware failure

16.7.8 AL-111 driver power supply voltage is too low

Possible Causes:

- (1) AC power input voltage is too low
- (2) Drive hardware failure

elimination method:

- (1) Check whether the AC power supply meets the drive specifications
- (2) If the above reasons are eliminated, the hardware may be faulty, please send it back to the dealer or the original factory for repair

16.7.9 AL-112 power line is not connected

Possible Causes:

- (1) The three-phase power line of the motor is off
- (2) The acceleration and deceleration time is set too short
- (3) Drive hardware abnormality

elimination method:

- (1) Check whether the UVW wiring from the driver to the motor is broken or loose
- (2) Increase the parameter Pn-307 (P6-11) maximum impulse (JERK)
- (3) Choose a driver with a larger rated voltage
- (4) Excluding the above reasons, there may be a hardware failure

16.7.10 AL-113 drive power off

Possible Causes:

- (1) Loose driver wiring
- (2) Abnormal power supply
- (3) Drive hardware abnormality

elimination method:

- (1) Check if the driver RST wiring is broken or loose
- (2) Check the power supply
- (3) Excluding the above reasons, there may be a hardware failure

16.7.11 AL-930 absolute encoder battery voltage is too low

Possible Causes:

- (1) The battery voltage is too low or there is no battery (3V)
- (2) Wiring error (the positive and negative poles of the battery are connected to the driver BAT+ BAT- respectively)
- (3) Parameter setting error

elimination method:

- (1) Measure the battery voltage and replace the battery
- (2) Check the wiring
- (3) If it is not an absolute encoder, please set the drive parameter Pn-904 to 0 and save and restart

16.7.12 AL-121 drive power module overcurrent

Possible Causes:

- (1) The UVW inside the motor is short-circuited or short-circuited to ground
- (2) Power line UVW is short-circuited or short-circuited to ground
- (3) The driver connection terminal UVW is short-circuited or short-circuited to ground
- (4) Mechanical factors cause the motor to stall
- (5) The drive power module is damaged
- (6) The three-phase resistance of the motor is unbalanced

elimination method:

- (1) The internal insulation of the motor may be poor, replace the motor
- (2) The power line may be short-circuited, replace the power line
- (3) The drive may be faulty, replace the drive
- (4) Eliminate mechanical factors, extend acceleration and deceleration time, and reduce load

16.7.13 AL-810 absolute encoder position is missing

Possible Causes:

- (1) The battery voltage is too low or there is no battery (2.5V)
- (2) Parameter setting error
- (3) Encoder is abnormal
- (4) The drive is abnormal
- (5) Encoder wire failure

elimination method:

- (1) Replace the battery
- (2) Replace the battery and restart the system
- (3) Replace the battery, set the parameter Pn-F44 to 1 and restart the drive
- (4) If it is not an absolute encoder, please set the drive parameter Pn-904 to 0 and save and restart
- (5) Cross test, check wire, encoder, driver problems

16.8 Encoder alarm

16.8.1 Encoder communication type setting error

Possible Causes:

- (1) The Pn-900 encoder communication interface type setting does not match the connected encoder
- (2) Check whether the encoder wire has poor grounding or broken skin
- (3) Check whether the encoder interface number set by Pn-901 is correct

elimination method

- (1) Restart the power after setting Pn-900 correctly
- (2) Reorganize the wire and confirm that the interference problem is solved and restart the power
- (3) Set the parameters correctly and restart the drive
- (4) Upgrade the driver software version

16.8.2 AL-304 encoder without feedback

Possible Causes:

- (1) Loose encoder wiring
- (2) Encoder communication interface type setting error
- (3) The encoder interface number is set incorrectly
- (4) Wire failure (short circuit, broken skin)
- (5) Encoder failure
- (6) The front plate of the driver is faulty
- (7) Encoder communication frequency is not supported
- (8) Encoder firmware update failed

elimination method:

- (1) Check whether the encoder wiring and pin definitions are correct
- (2) Set the parameters correctly and restart the drive
- (3) Replace the encoder wire
- (4) Replace the motor
- (5) Replace the drive

16.8.3 Encoder status is seriously abnormal

Possible Causes:

- (1) Encoder communication is disturbed
- (2) Poor grounding of the encoder wire shield
- (3) The encoder connector is in poor contact, loose or not installed
- (4) The communication interface type setting is wrong
- (5) Encoder firmware and hardware abnormalities

elimination method:

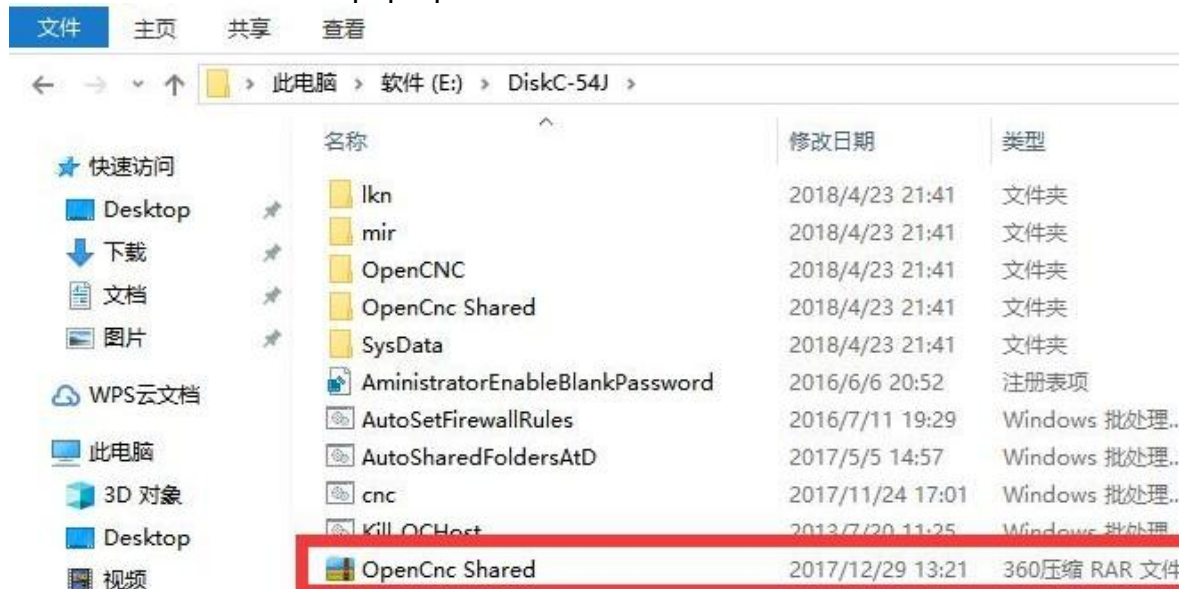
- (1) Check whether the drive status variables Pn-D73~Pn-D76 (D1-28, D1-29, D1-46, D1-47) have values
- (2) Check whether the encoder, motor, and machine are grounded

- (3) Check the connection between the encoder and the drive, the connection between the encoder and the motor, and whether the welding wire is correct
- (4) Detect whether the metal part of the first encoder interface at the driver end is connected to the driver housing.
- (5) Put magnetic rings on both sides of the encoder wire.
- (6) The encoder wiring should avoid running with the motor power line or other large power lines as much as possible
- (7) Correctly set the parameter Pn-900 (P3-20) encoder communication interface type, and restart the drive

16.9 Communication file recovery method

16.9.1 Processing power failure repair method

- (1) Open the computer→D drive→DiskC-(version number)→as shown below
- (2) Unzip the compressed package file "OpenCnc Shared" to the current folder, and select "Yes to all" in the pop-up window



16.9.2 HHcnc appears to open the pop-up window "Close Program"

Use the power-off repair tool to repair it

The lower machine alarms "Processing is interrupted, please re-check the processing data"

If this alarm occurs on the machine, you need to restart the machine normally once.

16.10 Heat engine (warming up) function use

Open "HHCNC" → menu bar "Machine Settings" → "Warm-up Program" (as shown in the figure below, applicable to 1S version)



The warm-up process will last for about 1 minute. After the operation is completed, please check whether all the drills have been retracted in place, and confirm that all the tools can be retracted for processing directly.

Note: This function must be operated after the machine is turned on normally.

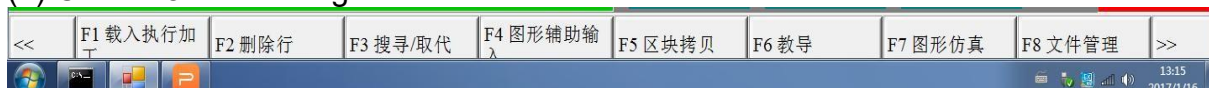
16.11 o010000 File selection

Note: There are no steps to solve the file that appears after sudden power failure, continuous startup of the machine, and complete password expiration.

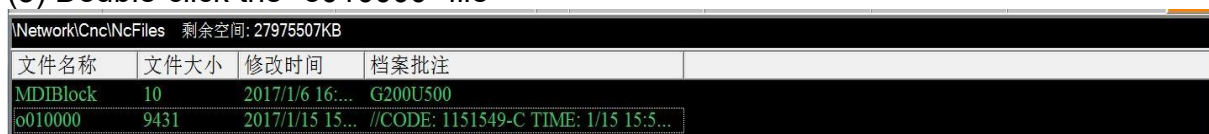
(1) Click "F2 Program Edit"



(2) Click "F8 File Management"



(3) Double-click the "o010000" file



(4) Click "F1 to load and execute processing"



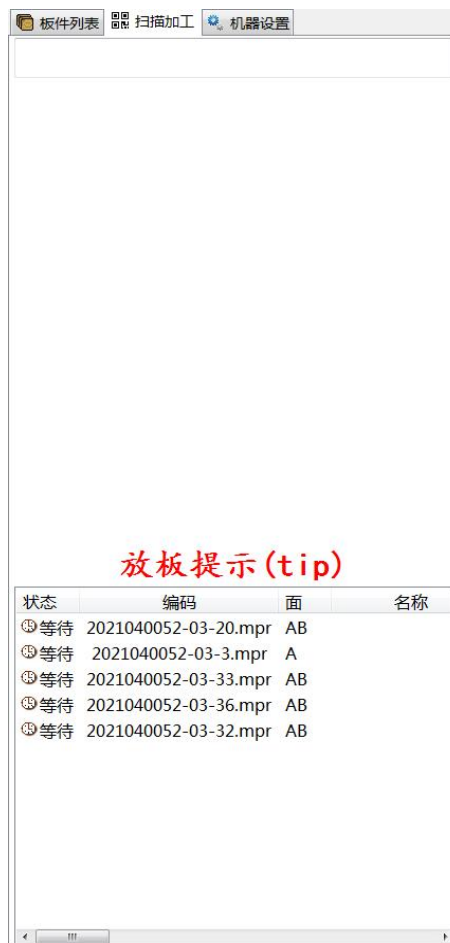
After the above operations are completed, it can be used normally

16.12 Continuous scanning processing

1). Click "Machine Settings" → tick "Scan multiple panels before automatic processing"



(2) Import the order number → scan processing, scan the information of two pieces of plates, the corresponding data will be displayed in the box, showing the processing status and coding information





(1) Click  The button "Generate CNC Code" triggers the start;

(2) Next, the equipment is in the "processing" state;

(3) Press the green start button for the first time → the machine tool is in the state of being processed, waiting for the board to be inserted;

(4) After the board is placed, press the green start button for the second time → the machine tool starts to process all the holes and slots of the board;

(5) After the processing is completed, take out the plate and wait for 1 second, the machine tool will automatically execute the processing task of the next plate;

(6) Repeat steps 2.4-2.6 for continuous processing.

(7) During continuous processing, if it is necessary to switch back to independent plate processing or manually generate data, the state of the machine tool needs to be maintained; "Machine tool is ready to wait";

(8) During the processing, if the panel is already in the processing state, if the current

panel does not need to be processed, you can click the system reset  button to skip the current panel;

(9) Use the function of turning 180 degrees, turning the plate around the Y axis, and turning the plate around the X axis.

状态	编码
⌚ 等待	L90090
⌚ 等待	L90100
⌚ 等待	L90101
⌚ 等待	L90120

(1) Note: During the processing, there must be more than one piece of data waiting here until the processing is completed.

(2) If there is a fault or the machine is in a fault state, please "clear the list" first, and then proceed to the fault processing of the lower machine position;

状态	编码
⌚ 等待	L90090
⌚ 等待	L90100
⌚ 等待	L90101
⌚ 等待	L90120

移动到最前
 移动到最后
 删除
 清空列表

(3) To use this function, you must be proficient in scanning and processing single plates;

(4) If a new user is using it, the data must be stable;

(5) Now supports 5+1 bar code format, that is, one code on five sides and one code on six sides.

16.13 Machine data backup

Open the computer→D drive→DiskC-(version number), compress the folder "DiskC-(version number)" and back it up. \ The folder "DiskC-(version number)" contains

- (1) Communication file of lower computer simulator
- (2) Lower computer simulator display file
- (3) Machine tool magazine data-HHcnc software data

Note: The Windows system of the machine can be used for regular system backup through Ghost

17. Schematic diagram of VGA display port switching

Picture shows:



This operation should be used to switch the display of the lower computer to the monitor. If you need to operate, please connect the keyboard to the USB port on the driver's integrated host.

- (1) Motor control system upgrade
- (2) The o010000 file cannot be recognized after the permission has been modified
- (3) In this state, the progress bar in the background will stay at 40%. After entering, please use the keyboard and press the Esc button to make the progress bar complete to the main interface. Use the keyboard F shortcut to perform the following steps. There is a pop-up window, please press ESC to remove
- (4) Press F2 program edit→F8 file management (switch up and down to O010000 file)→Enter (Entel)→F1 to load and execute processing, connect all interfaces back and use it normally

18. Accuracy debugging instructions

18.1 Preliminary accuracy adjustment

(1) Tool library parameter sequence adjustment items (X offset (front and back), Y (left and right), and Z (depth)). After adjustment, please synchronize the corresponding (+X) and (Y offset);

(2) Please reduce the speed to 30% for the first accuracy adjustment;

(3) After all the drill bits are installed, all the tools are called out to ensure that all the tool tips are on the same plane ($\pm 1\text{mm}$);

(4) When the Z axis moves to the origin position, the horizontal tool extends out of the two manipulators and moves forward and backward without collision;

(5) Adjust the V axis (moving the drill bag left and right) and the manipulator (X axis and U axis) without collision when moving back and forth (you can manually push the operation after pressing the emergency stop), and set the positive and negative limits of the machine Value, to ensure that there can be no collision in the state of the mechanical software limit;

(6) Parameter No. 124 of the position point of the positioning column Z (initialize the designated position), this parameter needs to be re-calibrated after the accuracy of the upper drilling package reference tool has been debugged;

(7) Adjust the distance of the reference positioning column in the Y direction, configure tools→pneumatic components→positioning column base point Y (the smaller the value, the leftward);

Note:

A: Please process a plate with a plate width of 100 to ensure that there is no collision with the back and forth movement of the manipulator when the positioning column comes down;

B: This parameter needs to be re-calibrated after the accuracy of the benchmark tool of the upper drilling package has been debugged.

Lean to the side of the board (the larger the value, the greater the space); the compression position (the larger the value, the more left the measurement is)-the conventional parameter No. 129 compression position can also be adjusted during installation and debugging Little Yue goes to the right).

Note: Upper drill package reference tool T1
Drilling bag reference tool

18.2 T21 Precision adjustment of the datum tool of the upper drilling package (T1)

18.2.1 Vertical hole precision debugging

(1) Fore and aft distance-configuration tool → pneumatic components → positioning column base point X (the larger the value, the smaller the distance from the hole to the datum)

(2) Left and right distance-new-generation G54 processing point (the smaller the value, the smaller the distance from the hole to the datum) Y

Note: After adjusting this parameter, you need to pay attention to the positive and negative limits of the Y-axis in the upper machine position (HHCNC) and need to be re-calibrated

(3) Up and down direction-new-generation G54 processing point (the larger the value, the higher the drill bit is-the hole is shallower) Z

Note: After adjusting this parameter, you need to pay attention to the positive and negative limits of the Z axis in the upper machine position (HHCNC) need to be re-calibrated

18.2.2 Horizontal hole precision debugging

Note: When debugging, please process the left and right horizontal holes first, and adjust the front and back directions after the Z direction is adjusted. Horizontal tool 19-|20-processing the safe distance adjustment of the tool tip from the edge of the board
Configure tools→tool library→safety distance (the smaller the value, the closer to the edge of the board)

(1) Z direction-horizontal tool (the larger the Z value of the tool magazine, the closer the hole is to the table)

(2) X direction-horizontal tool No. 17+/18+ (the larger the value, the shallower the hole)

(3) X direction-horizontal tool No. 17-/18- (the larger the value, the deeper the hole)

(4) X direction-horizontal tool No. 19+/20+ (the greater the value, the greater the distance from the hole to the datum)

(5) X direction-horizontal tool No. 19-/20- (the greater the value, the greater the distance from the hole to the datum)

(6) Y direction-horizontal tool No. 17+/18+ (the larger the value, the more the hole goes to the left)

(7) Y direction-horizontal tool No. 17-/18- (the larger the value, the more the hole goes to the left)

(8) Y direction-horizontal tool No. 19+/20+ (the larger the value, the deeper the hole)

(9) Y direction-horizontal tool No. 19-/20- (the larger the value, the shallower the hole)

18.2.3 Precision adjustment of upper milling cutter

(1) Front and rear direction (the greater the value, the greater the distance from the slot to the reference) X

(2) Left and right direction (the greater the value, the greater the distance from the slot to the reference) Y

(3) Depth direction (the larger the value, the deeper the groove) Z

18.3 Precision adjustment of the benchmark tool of the drill-down package (T21)

18.3.1 Accuracy adjustment of the datum cutter of the drill-down package

The distance between the reference tool of the lower drilling package and the reference tool of the upper drilling package:

- (1) Front-to-back distance Configuration tool→Common settings→No. 121 (the smaller the value, the smaller the distance from the hole to the reference) X;
- (2) Left and right distance-----new-generation G54 processing point (the smaller the value, the smaller the distance from the hole to the datum) V (need to adjust the positive and negative travel limits of the V axis);
- (3) Up and down direction New generation G54 processing point (the larger the value, the more the drill goes up-that is, the hole depth) W.

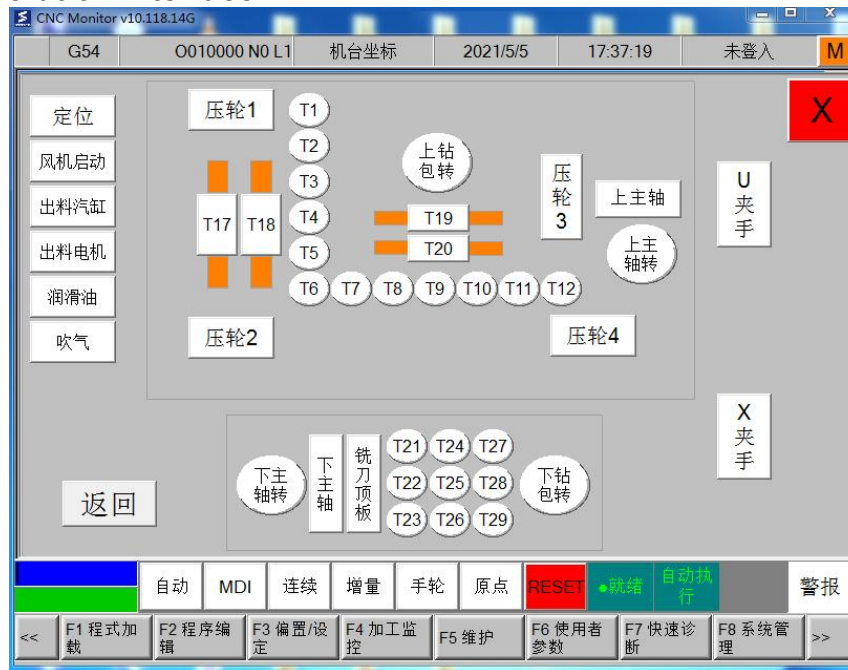
Note: To adjust the overall accuracy of the drill package, you need to adjust the left and right processing points of the drill package. The horizontal slotting V axis needs to move to the negative limit of the V axis. When debugging, pay attention to moving the manipulator back and forth when the V axis moves to the negative limit. Ensure that the collision safety distance is greater than 1mm.

18.3.2 Accuracy adjustment of bottom milling cutter

Configuration tool→tool magazine W1

- (1) Fore and aft direction (the greater the value, the greater the distance from the slot to the reference point) X offset
- (2) Left and right direction (the greater the value, the greater the distance from the slot to the reference) Y
- (3) Depth direction (the larger the value, the deeper the groove) Z

18.4 Manual operation interface



Software control "M button" (the following operations are all carried out without alarm)

- (1) Including the independent control of the entire machine's actions. Please switch to the "continuous" mode in the current interface. After clicking all the above independent buttons, the corresponding hardware actions will be executed, and each axis can be continuously moved according to the machine coordinate system (The mode is as shown



- (2) To return to the origin of each axis, please switch the mode "origin" (the mode is as shown in the figure below



- (3) Operation precautions

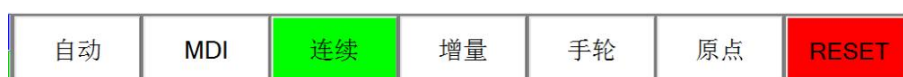
A: Please make sure that all axes are in an independent and safe state before using this function

B: When returning to the origin independently, please follow the order

Z→W→A→V→Y→X→U to return to the origin

C: Please confirm that there are no obstacles on the machine when returning to the origin with one key

- (4) Please use "automatic" mode for processing



Tool box list			Packing list
			Cnc drilling machine
No.	name	Quantity	Remarks
1	Combination wrench	8 pieces	21、12-12、13-13
			16-16、17-17、18-18
			19-19、21、27-30
2	Allen wrench	1 set	
3	Milling cutter wrench	1 piece	
5	12 Adjustable wrench	1 piece	
6	Phillips screwdriver	1 piece	
7	Standard drill	More	
8	Machine foot adjustment pad	8	
9	Scan gun	2	
10	butter	1 bottle	
11	Grease gun	1 piece	
12	Certificate of conformity	1 serving	
13	Toolbox	1 piece	
14	user's manual	1 serving	
Note: (Button switch; Scanner gun; Proximity switch) are vulnerable parts and are not covered by the warranty			

19. Warranty regulations

(1) The warranty period of this product is one year.

(2) During the warranty period, faults arising from normal use according to the instruction manual will be repaired free of charge.

(3) During the warranty period, if one of the following occurs, it must be repaired for a fee:

A. It is not possible to provide this guarantee and valid proof of purchase.

B. Failure and damage caused by misuse and improper repair.

C. Failure and damage caused by the user during transportation, moving and falling.

D. Failure and damage caused by other unavoidable external factors.

E. Improper use causes damage to the equipment caused by water or other solutions.

F. The damage caused by the power supply and voltage in the use guide.

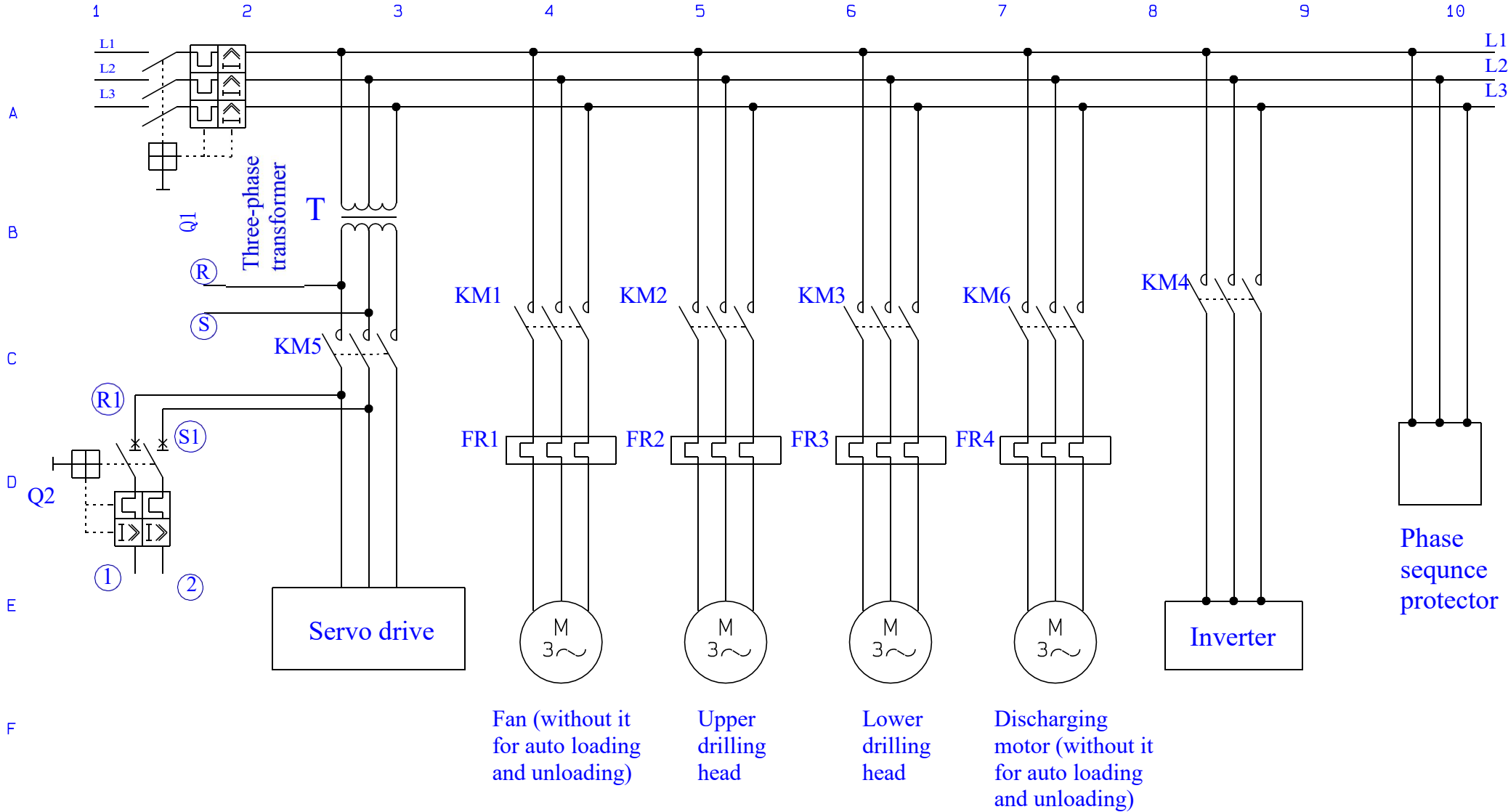
G. Damage caused by man-made operations that do not conform to the use of equipment.

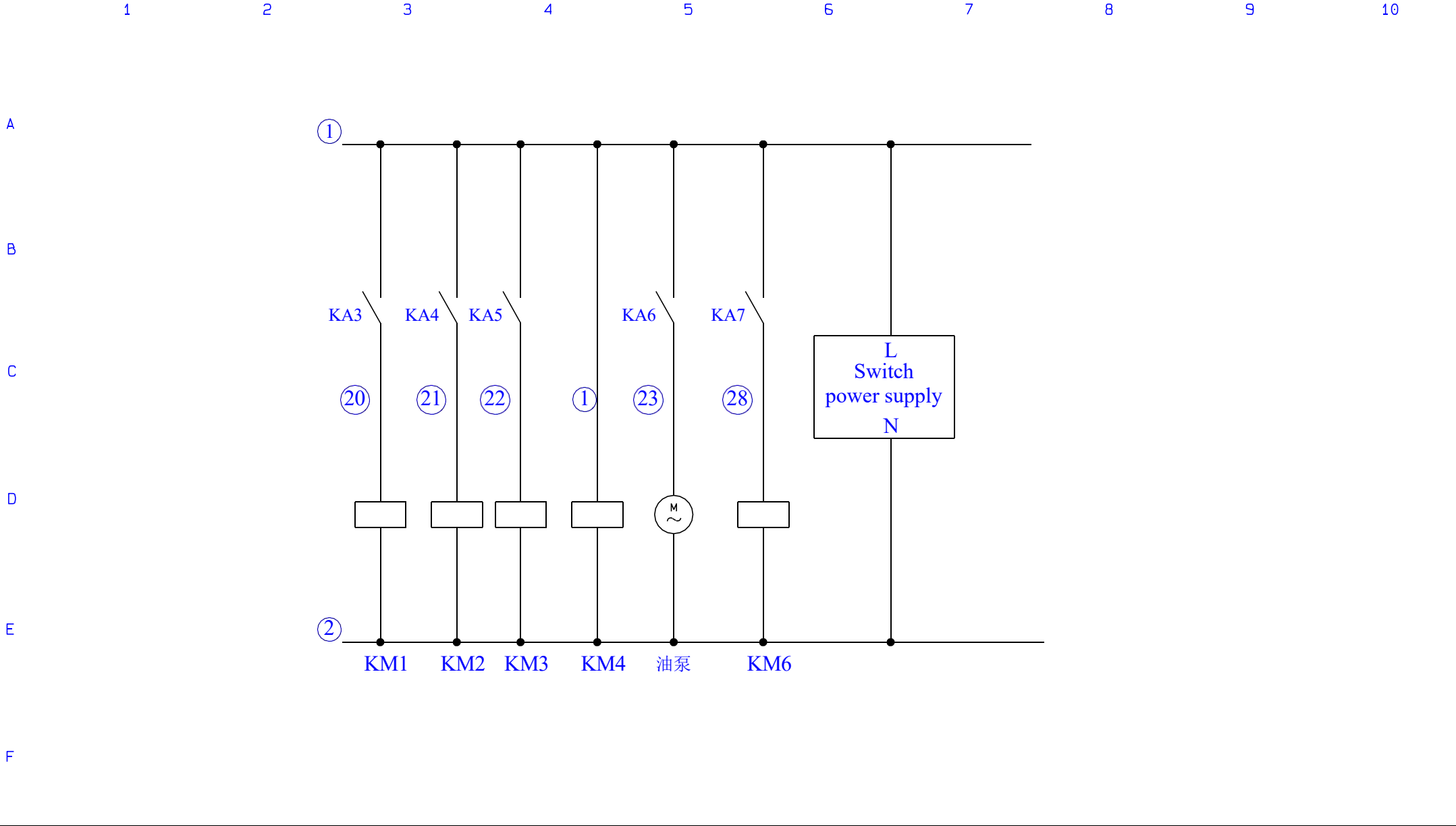
H. Loss of wearing parts.

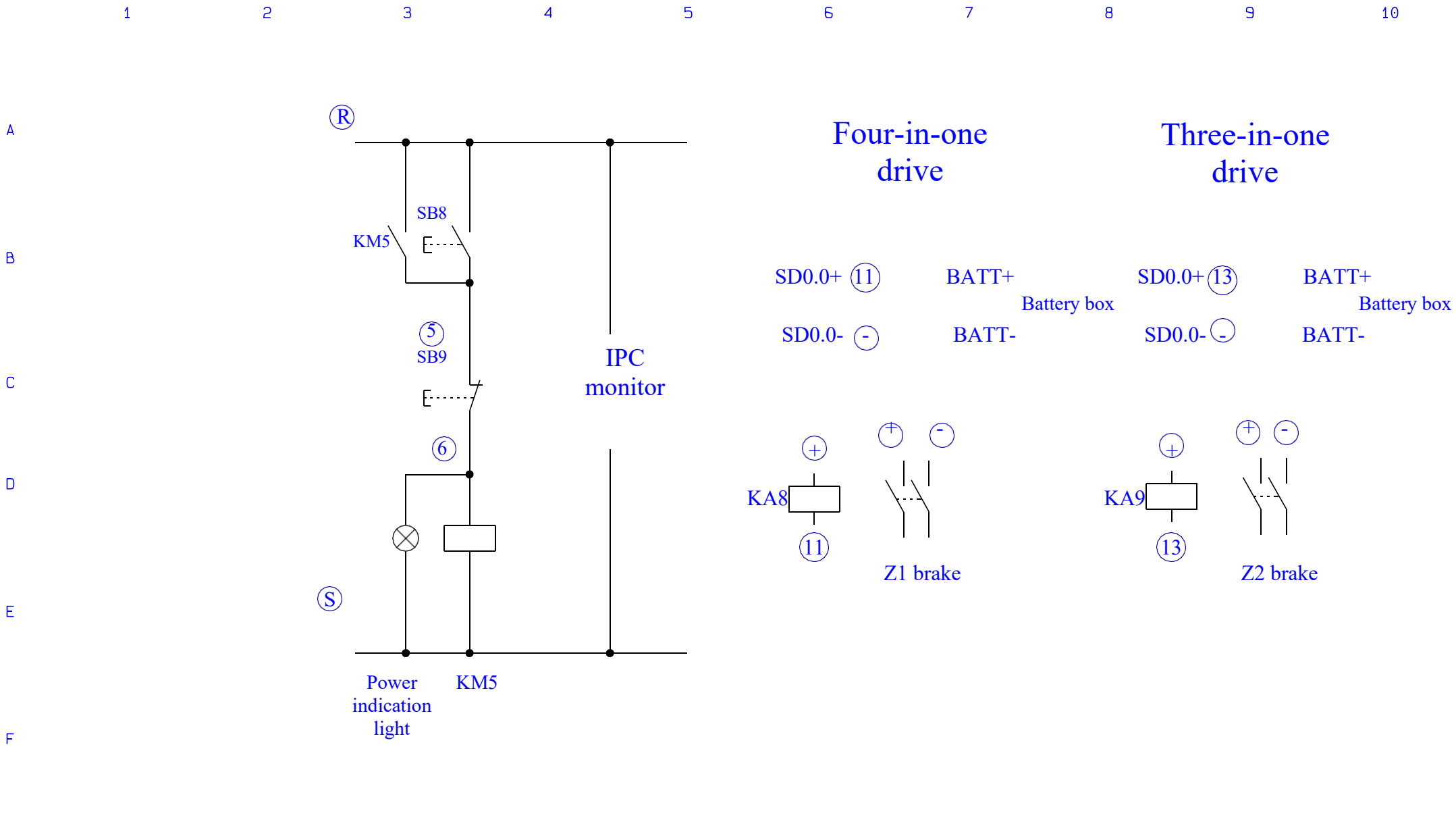
(4) Only the above guarantees are made, and no other express or implied guarantees (including implied guarantees of marketability, reasonableness and adaptability to a particular application and application) are not made, not in the contract, civil negligence , Or other aspects, the company is not responsible for any special, incidental or indirect damages.

Guangdong Shunde Changsheng Machinery Manufacturing Co., Ltd.

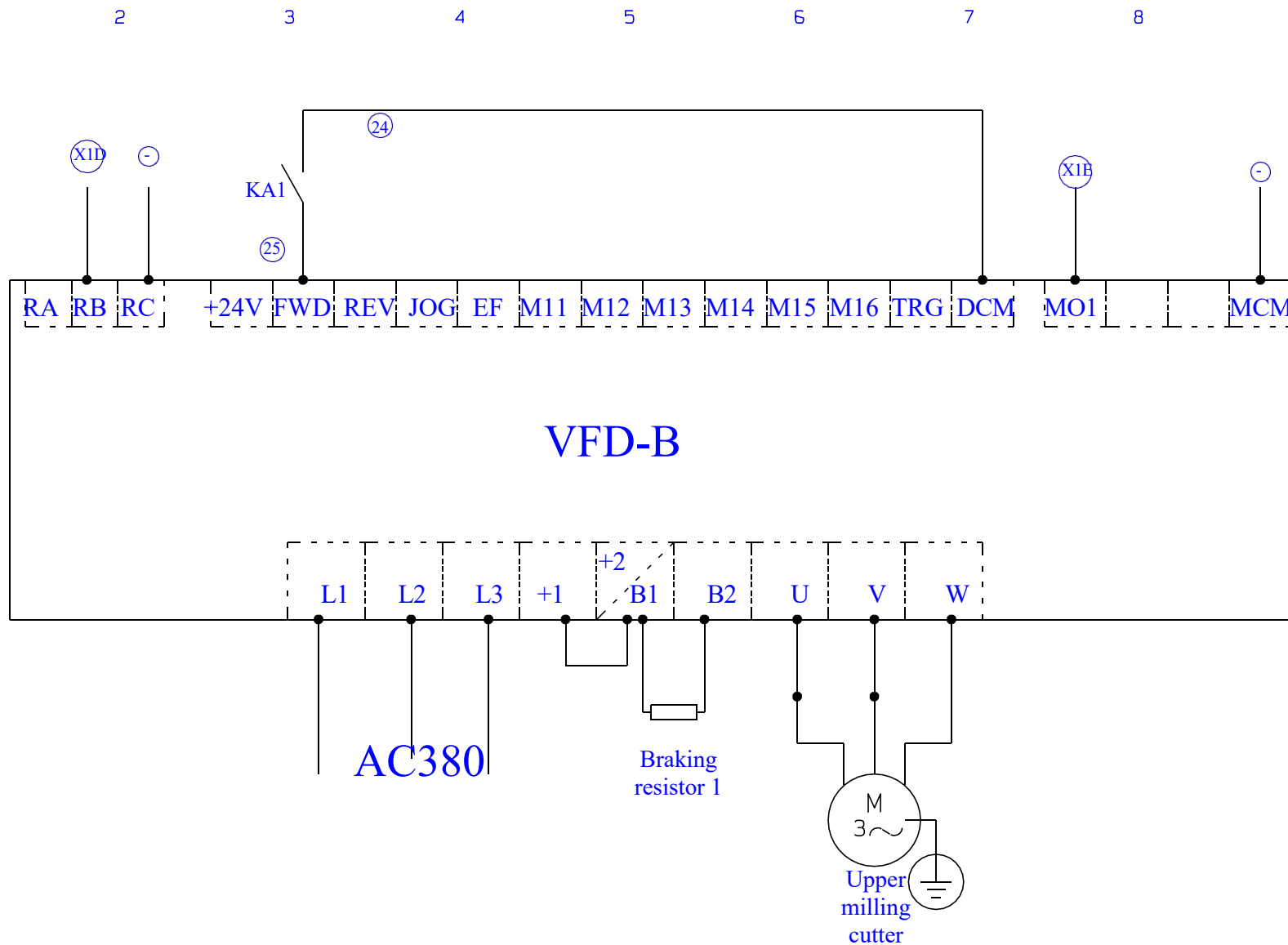
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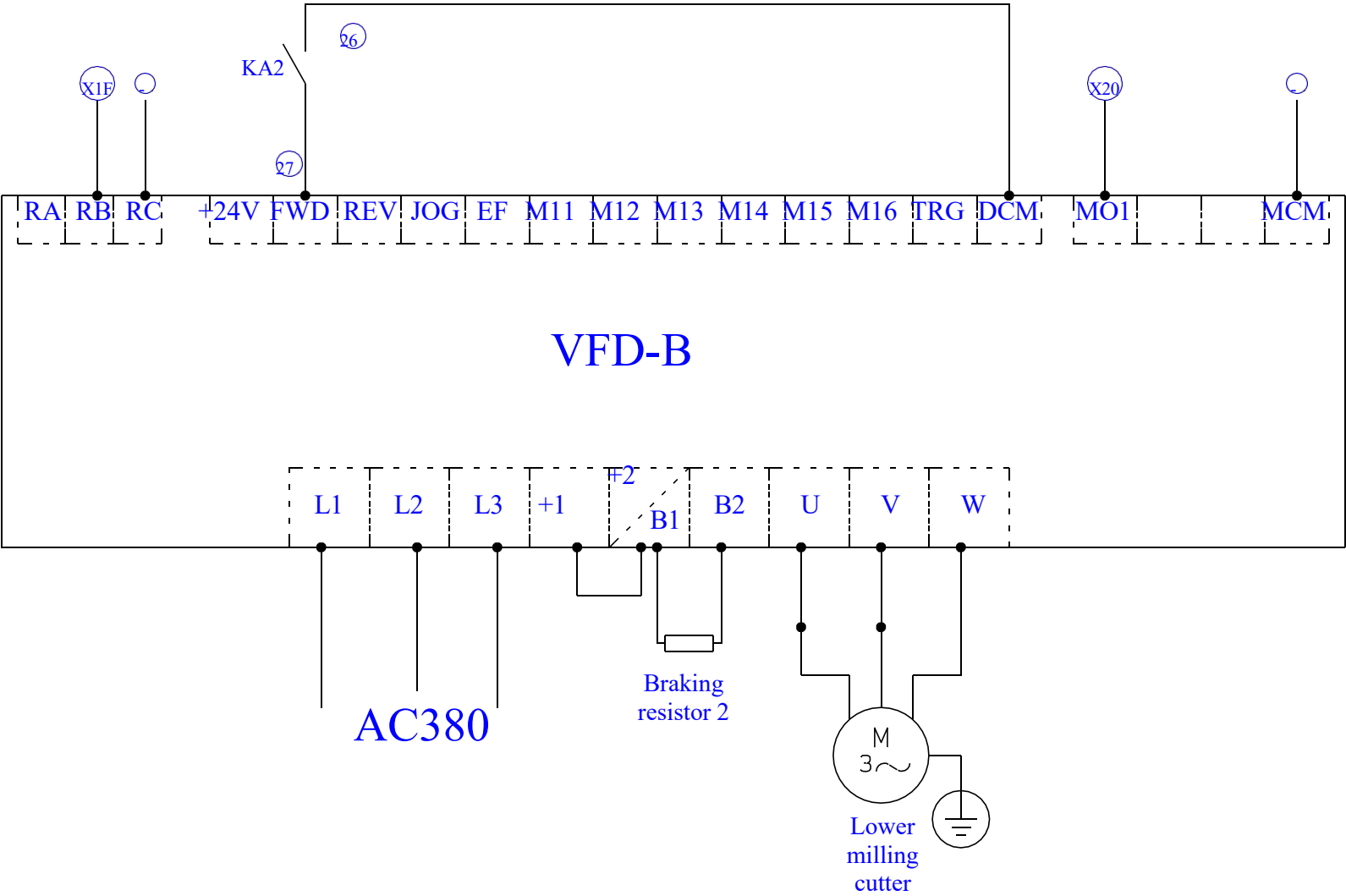


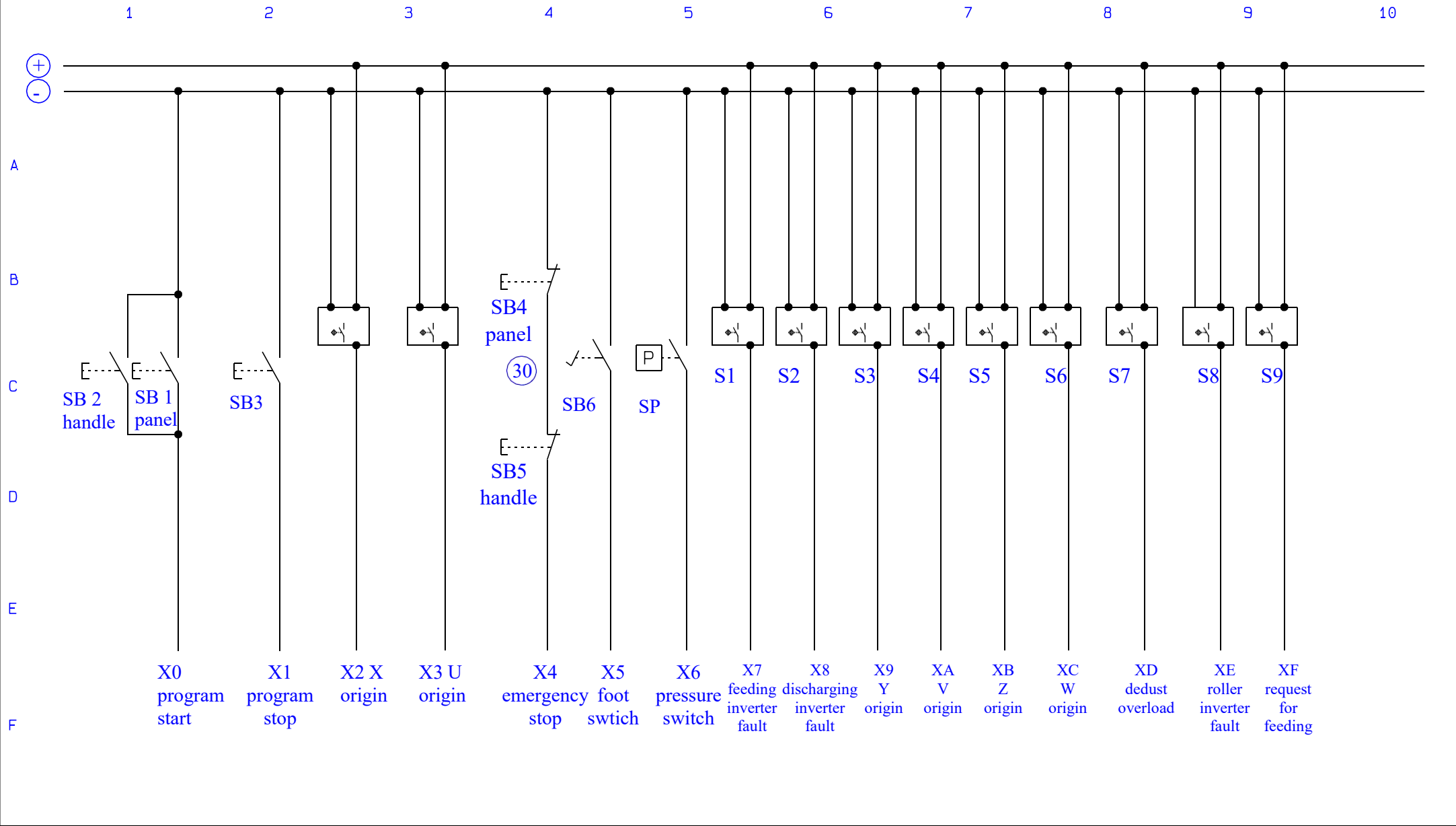


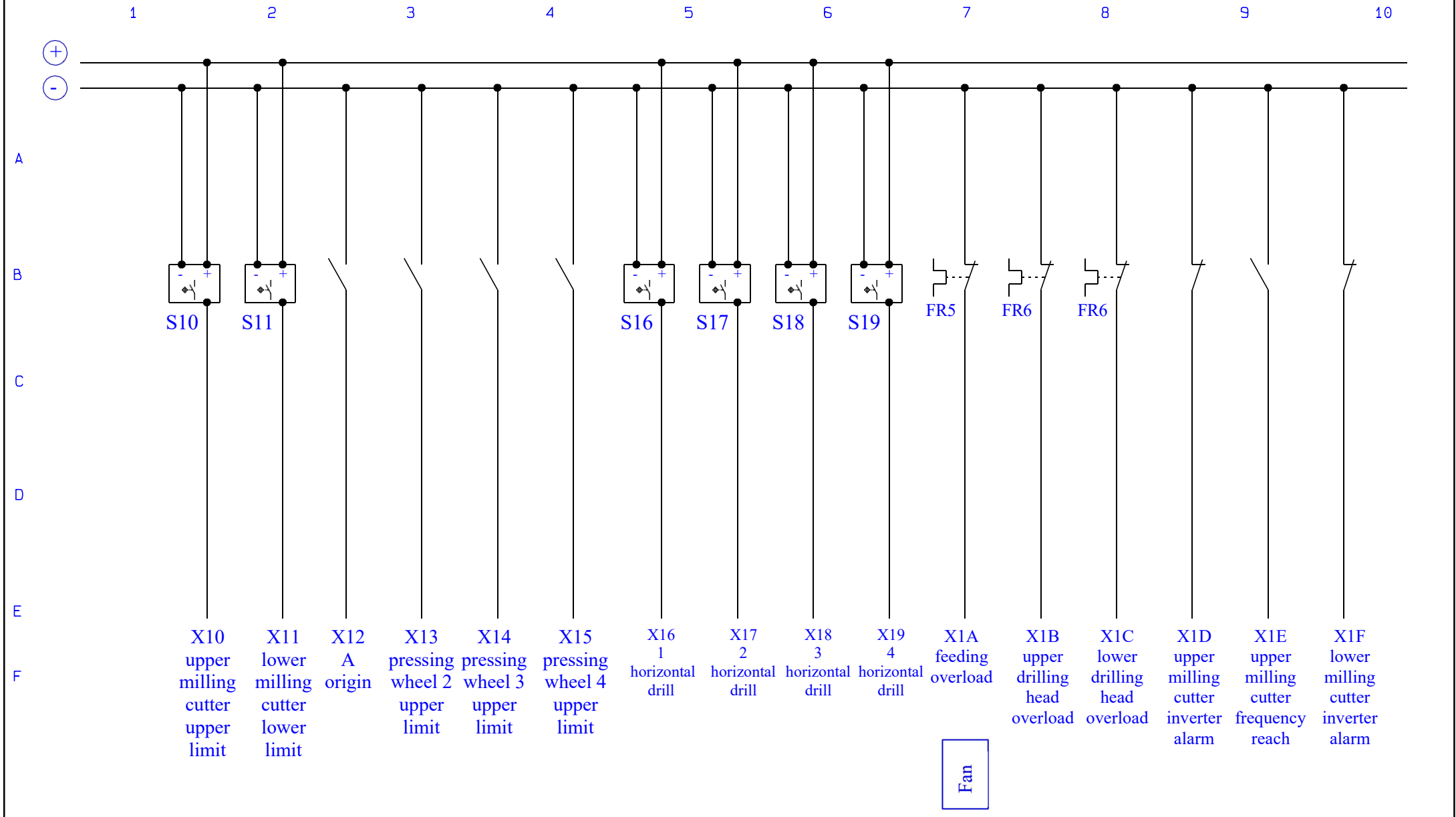


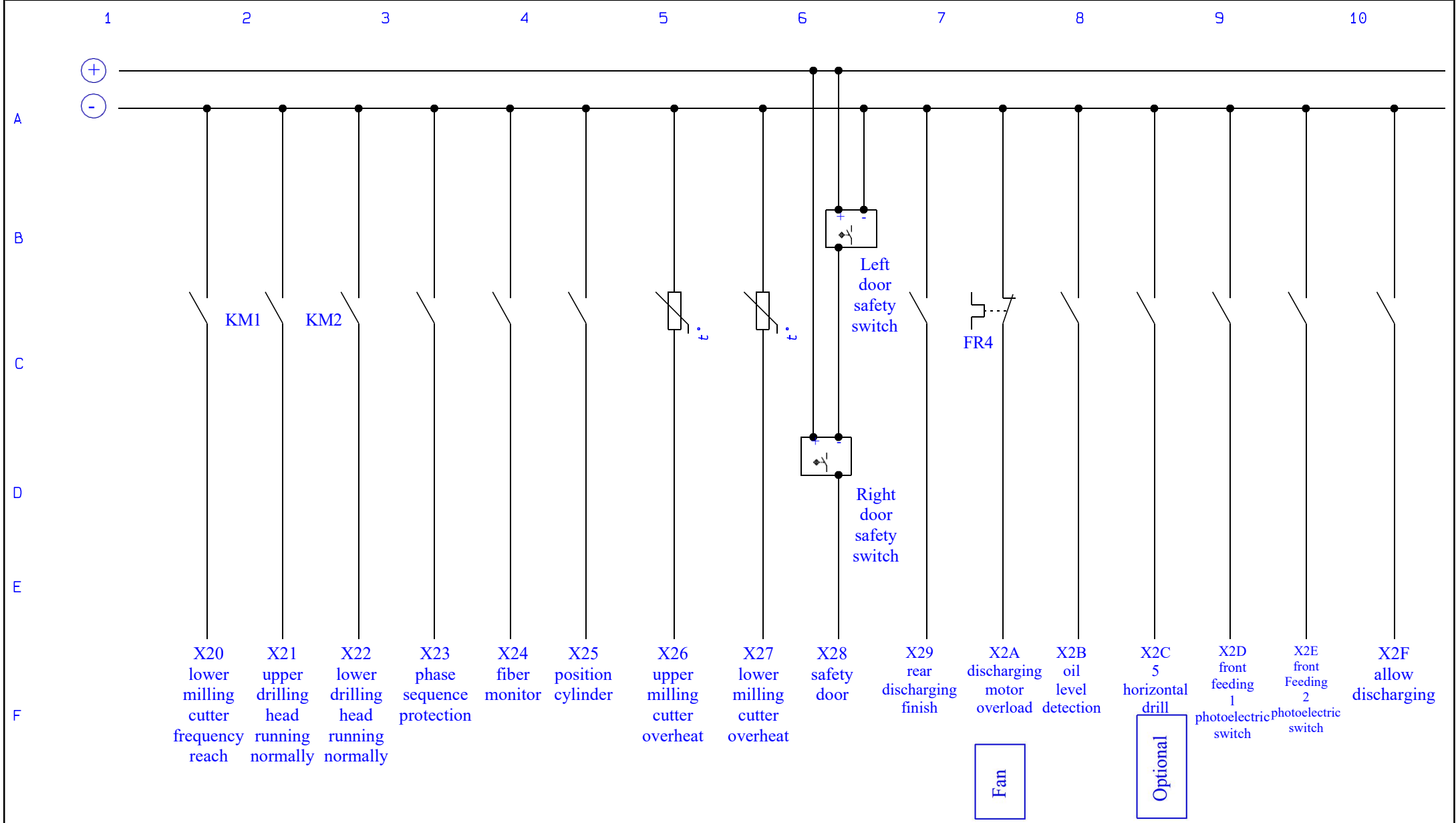
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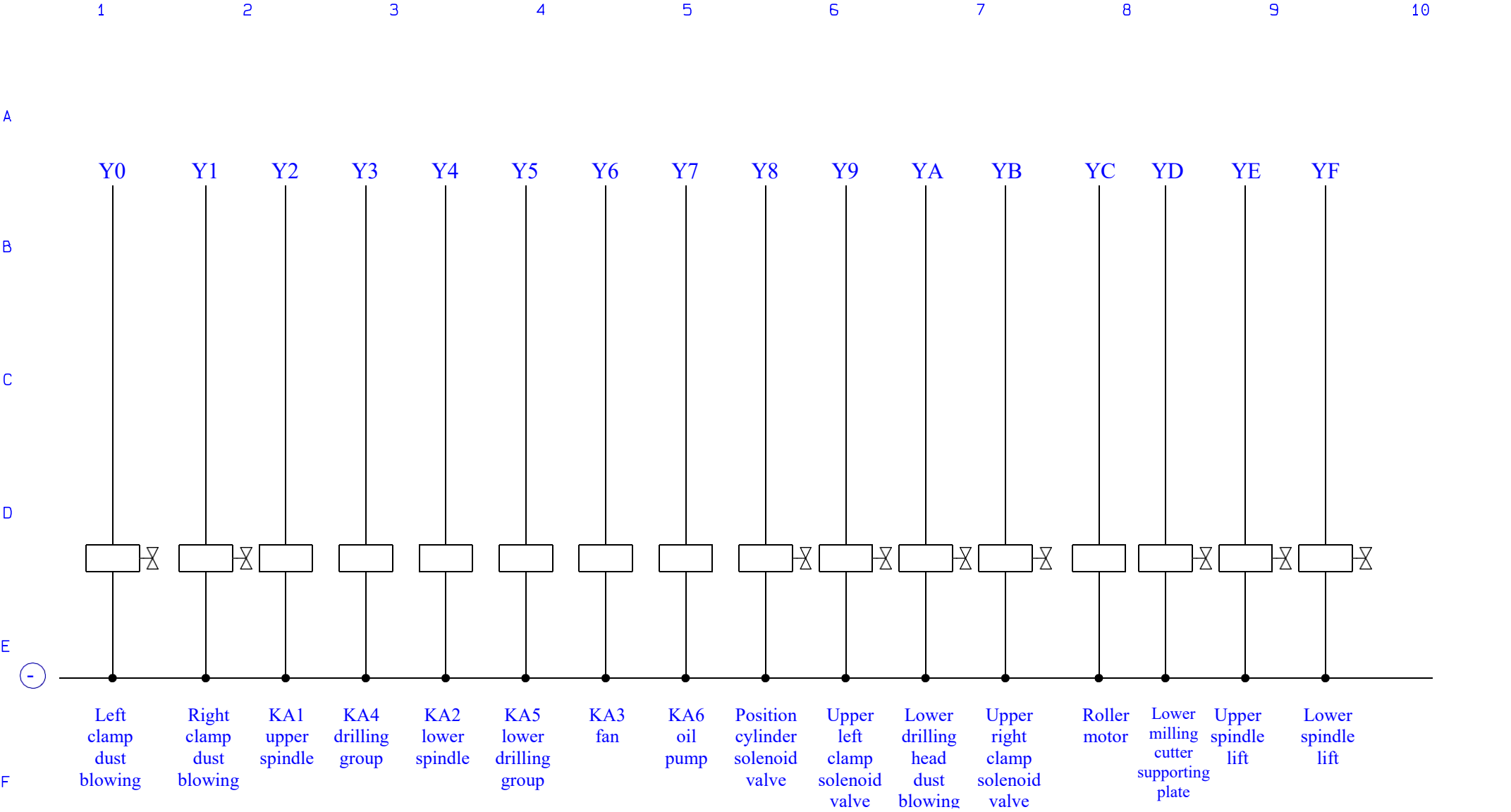




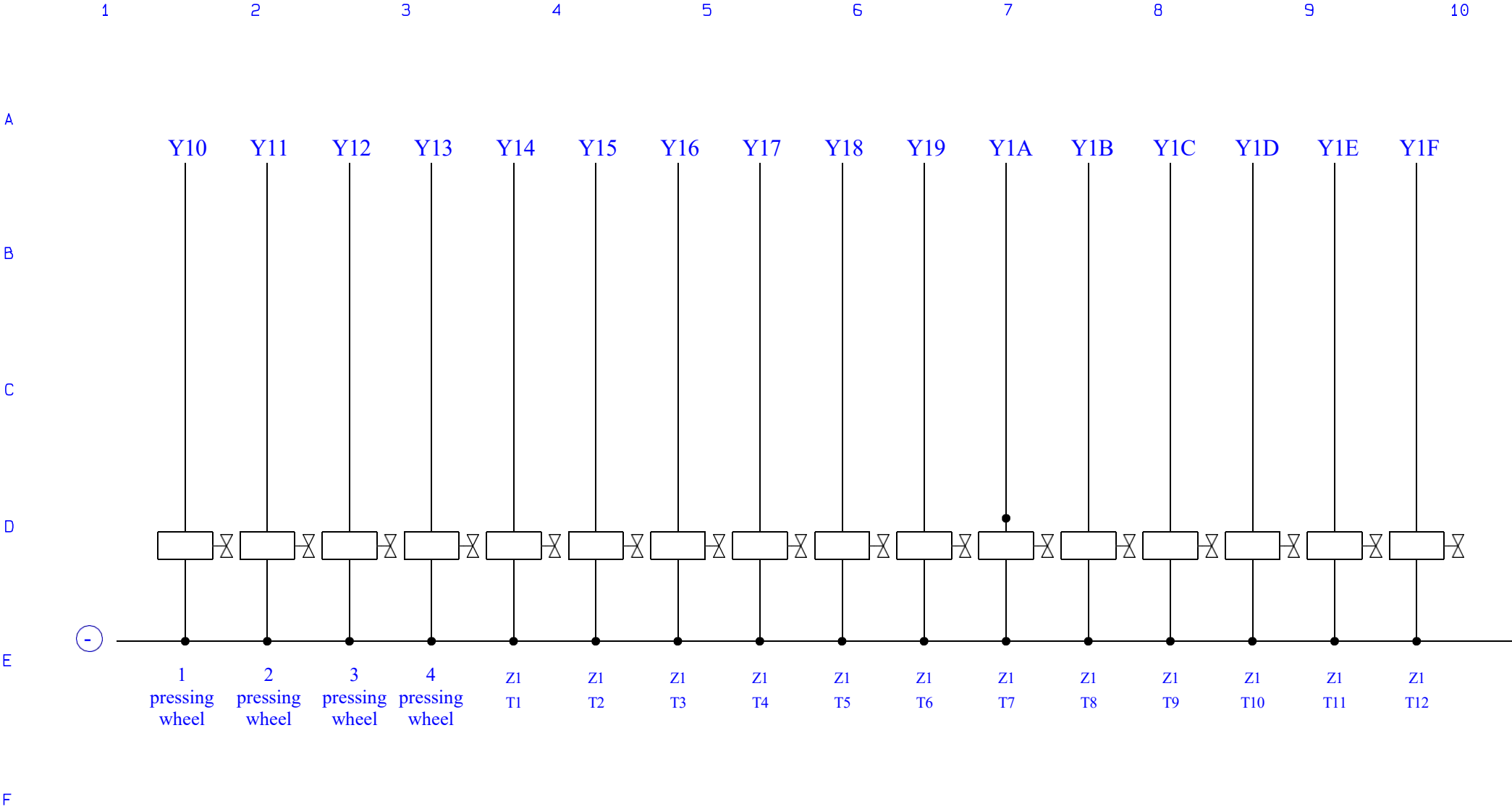




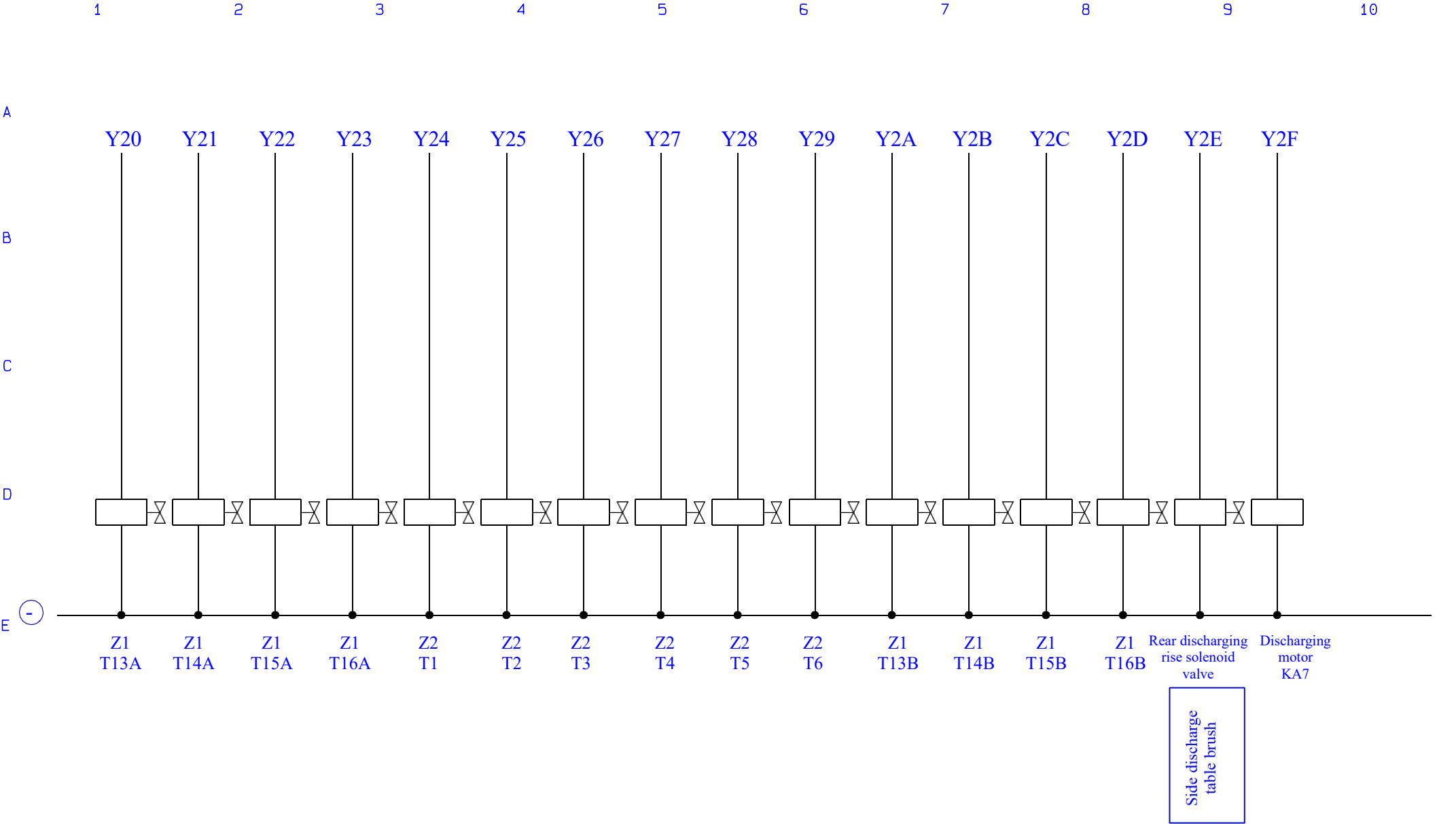
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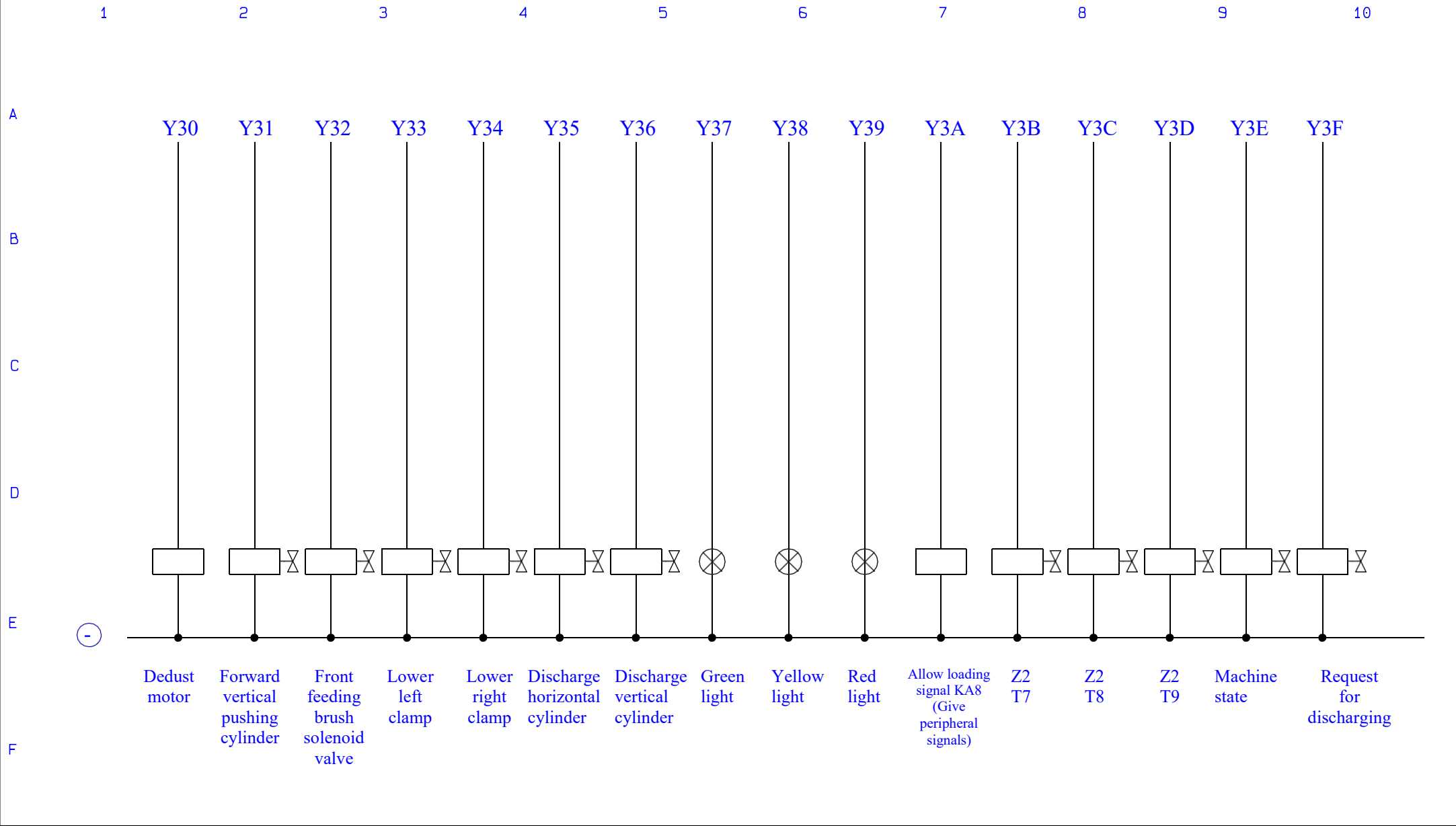
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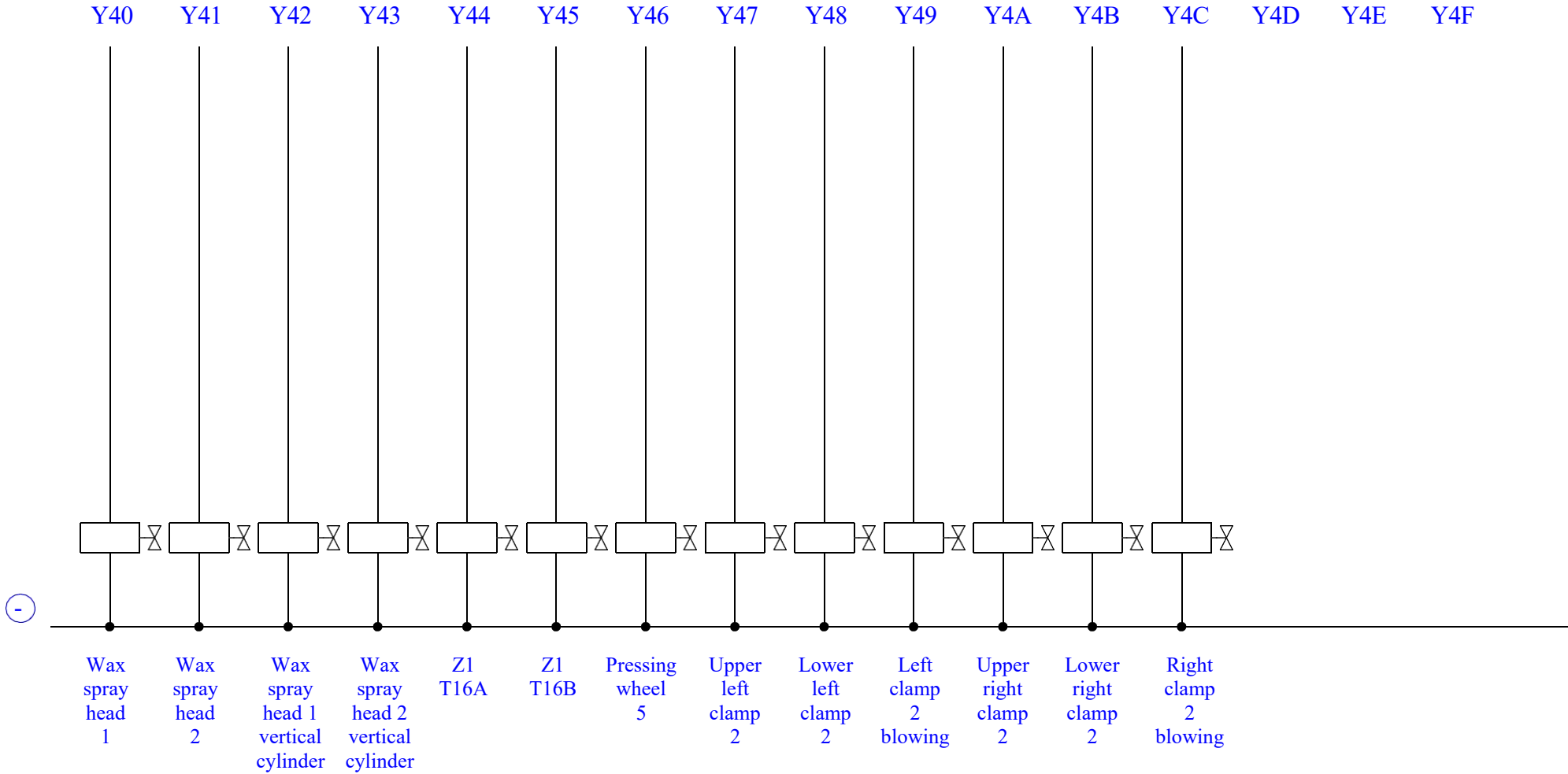
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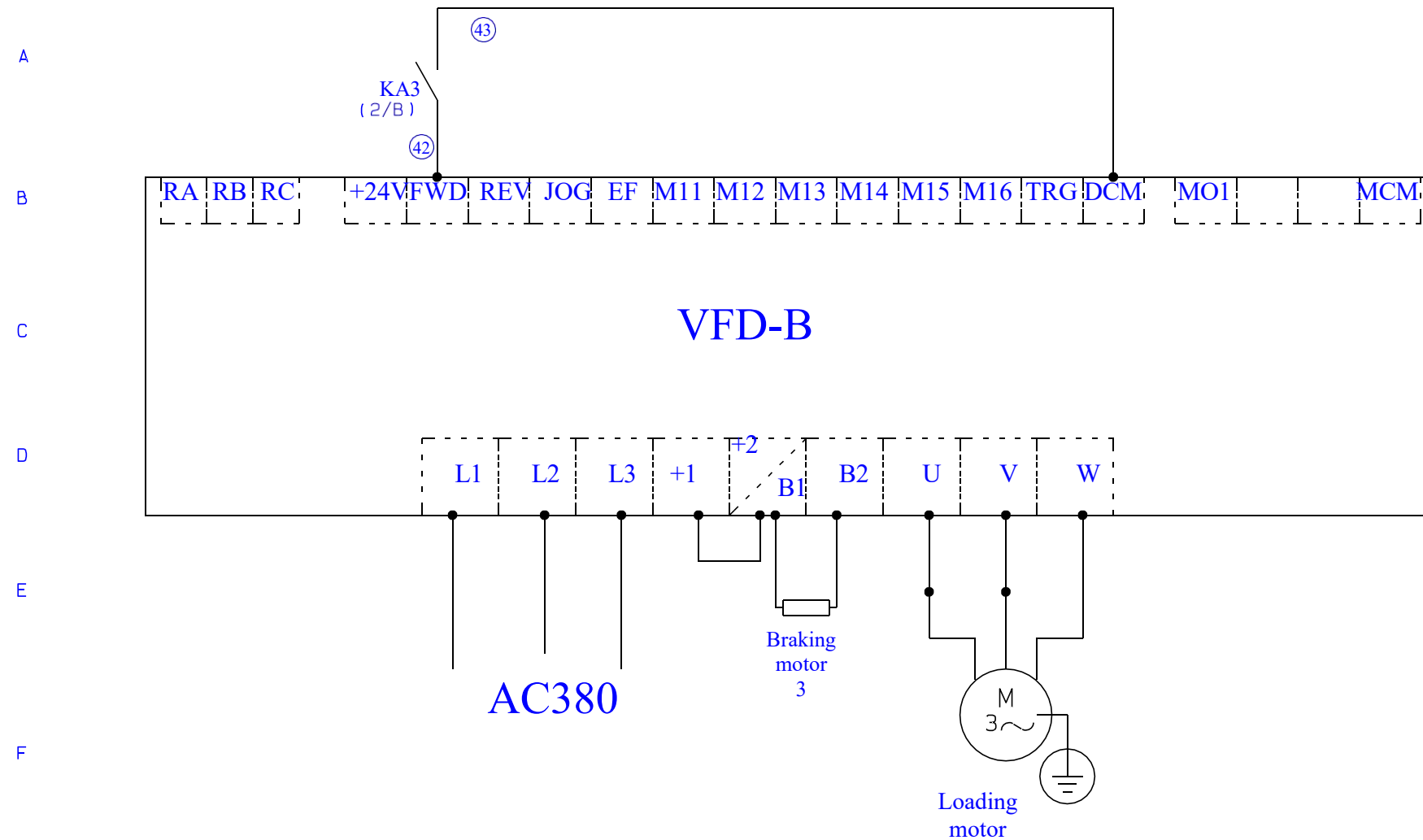


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